

DRIVELINE/AXLE

03
SECTION

TRANSFER 03-16
TECHNICAL DATA 03-50

SERVICE TOOLS 03-60

03

03-16 TRANSFER

TRANSFER CLEANING 03-16-1
TRANSFER DISASSEMBLY 03-16-2

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TRANSFER CLEANING

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Cleaning Precautions

1. Clean the surface of the transfer using steam and cleaning fluids when disassembly.

Warning

- Always wear safety glasses when using compressed air since the foreign material could be blown by the compression air and damage your eyes.
2. Clean removed components with cleaning fluids and use compressed air to blow off the oil. Clean the oil holes and passages with compressed air.

TRANSFER

TRANSFER DISASSEMBLY

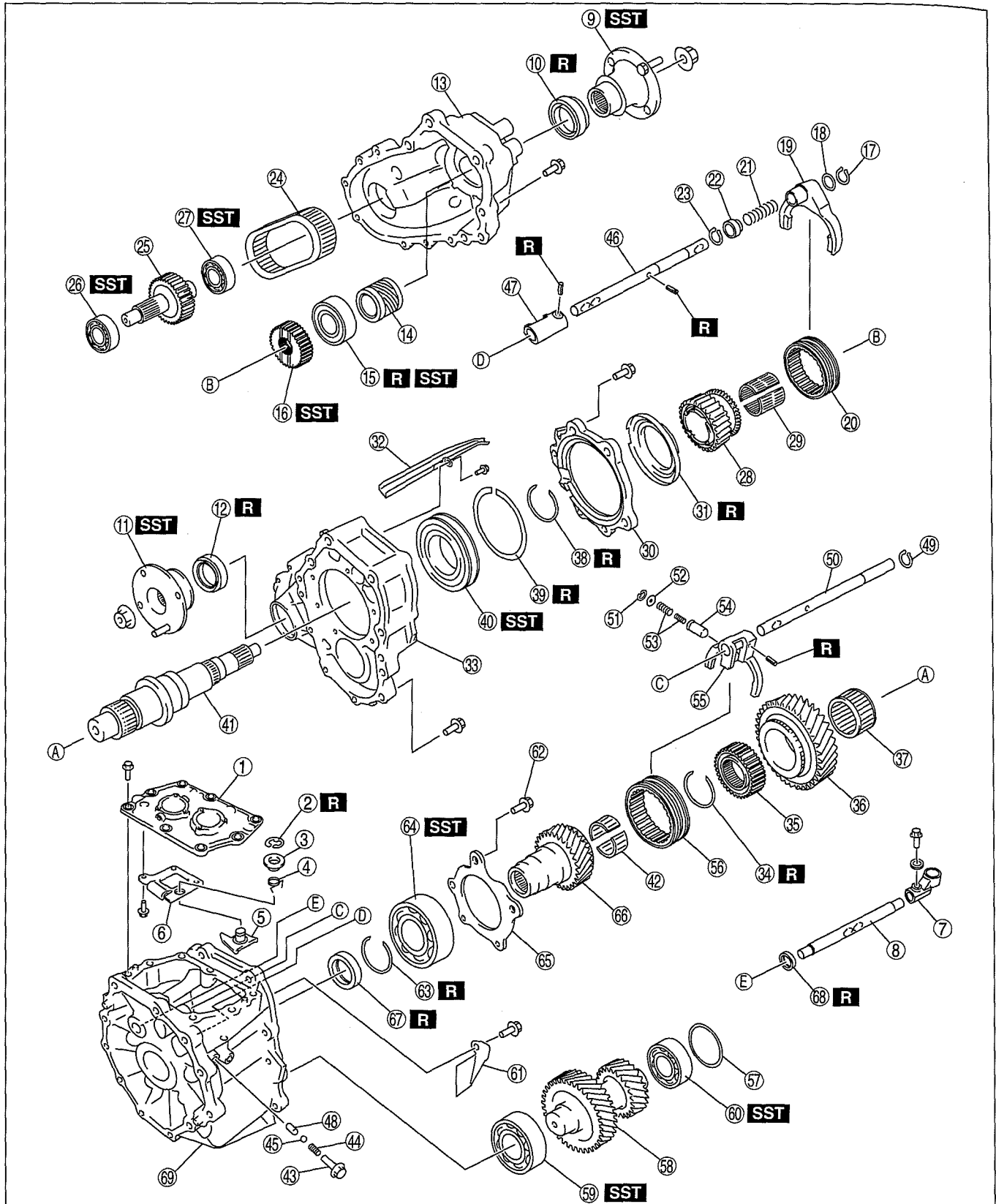
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Before Service Precautions

- To prevent foreign material from entering the transfer, perform disassembly and servicing in a clean, dust-free environment.
- Inspect the each part while disassembling.

Transfer Disassembly

1. Disassemble in the order shown in the figure.



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1	Control case
2	E-ring
3	Spacer
4	Reverse gate spring
5	Reverse gate lever
6	Guide plate
7	Control rod end
8	Control rod
9	Companion flange (rear case side) (See 03-16-4 Companion Flange Disassembly Note.)
10	Oil seal (rear case)
11	Companion flange (center case side) (See 03-16-4 Companion Flange Disassembly Note.)
12	Oil seal (center case)
13	Rear case
14	Speed drive gear
15	Mainshaft rear bearing (See 03-16-5 Mainshaft Rear Bearing Disassembly Note.)
16	2W/4W clutch hub (See 03-16-5 2W/4W Clutch Hub Disassembly Note.)
17	Snap ring
18	Spacer
19	2W/4W shift fork
20	2W/4W hub sleeve
21	Spring
22	Collar
23	Snap ring
24	Chain (See 03-16-6 Chain, Front Drive Shaft and Sprocket Disassembly Note.)
25	Front drive shaft (See 03-16-6 Chain, Front Drive Shaft and Sprocket Disassembly Note.)
26	Front drive shaft bearing (center case side) (See 03-16-6 Front Drive Shaft Bearing Disassembly Note.)
27	Front drive shaft bearing (rear case side) (See 03-16-6 Front Drive Shaft Bearing Disassembly Note.)
28	Sprocket (See 03-16-6 Chain, Front Drive Shaft and Sprocket Disassembly Note.)
29	Needle bearing
30	Center bearing retainer
31	Oil catcher
32	Oil passage
33	Center case

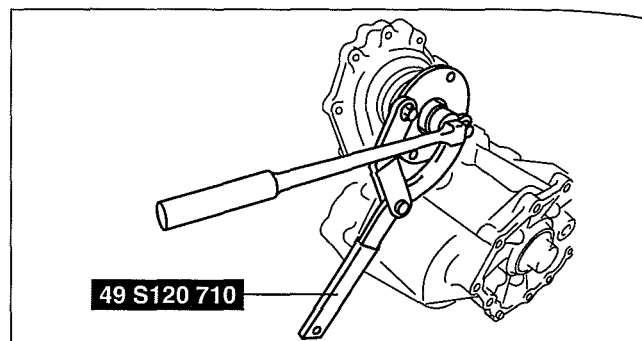
34	Retaining ring
35	H/L clutch hub
36	Low gear
37	Needle bearing
38	Retaining ring
39	Retaining ring
40	Mainshaft center bearing (See 03-16-7 Mainshaft Center Bearing Disassembly Note.)
41	Mainshaft
42	Needle bearing
43	Retaining bolt
44	Spring
45	Steel ball
46	2W/4W shift rod
47	2W/4W shift end
48	Interlock pin
49	Snap ring
50	H/L shift rod
51	Snap ring
52	Plain washer
53	Spring
54	Plunger
55	H/L shift fork
56	H/L hub sleeve
57	Adjustment shim
58	Countershaft
59	Countershaft front bearing (front case side) (See 03-16-7 Countershaft Bearing Disassembly Note.)
60	Countershaft rear bearing (center case side) (See 03-16-7 Countershaft Bearing Disassembly Note.)
61	Baffle plate
62	Retaining bolt
63	Retaining ring
64	Maindrive gear bearing (See 03-16-7 Maindrive Gear Bearing Disassembly Note.)
65	Retainer
66	Maindrive gear
67	Oil seal (control rod)
68	Oil seal (front case)
69	Front case

TRANSFER

Companion Flange Disassembly Note

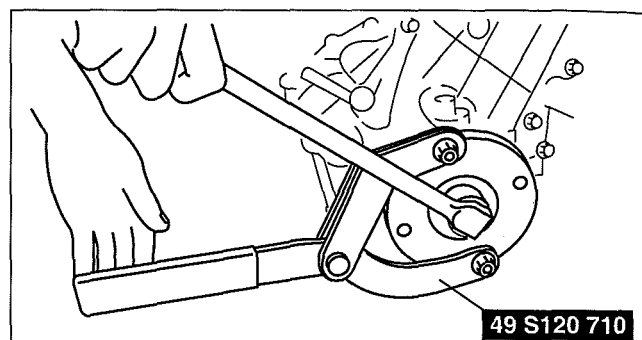
1. Hold the companion flange using the SST.

Center case side



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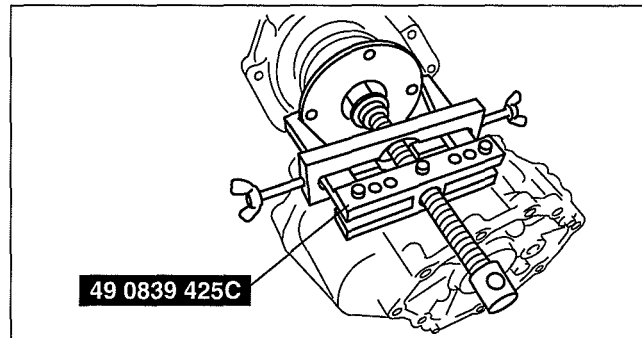
Rear case side



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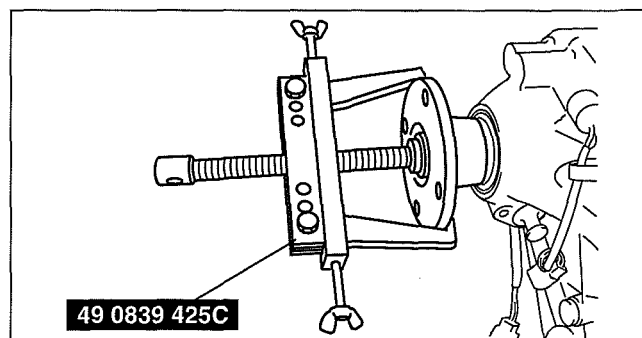
2. Remove the locknut.
3. Remove the companion flange using the SST.

Center case side



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Rear case side

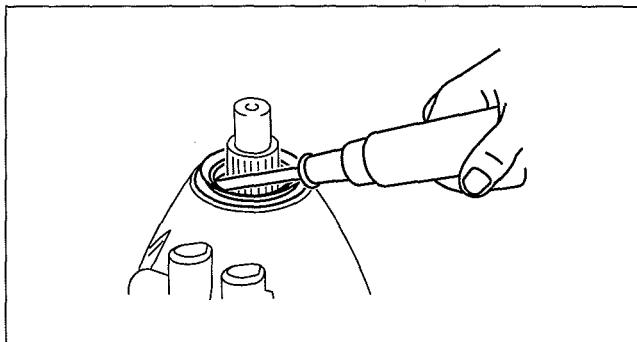


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Oil Seal (Rear Case) Disassembly Note

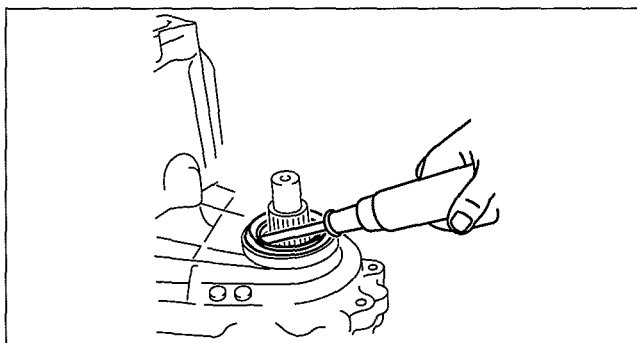
1. Remove the oil seal (rear case) using a flathead screwdriver as shown in the figure.



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Oil Seal (Center Case) Disassembly Note

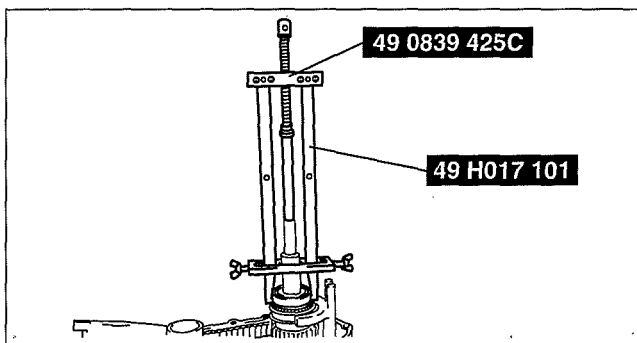
1. Remove the oil seal (center case) using a flathead screwdriver as shown in the figure.



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Mainshaft Rear Bearing Disassembly Note

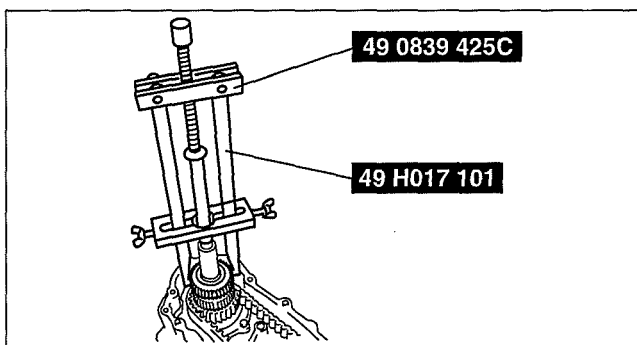
1. Remove the mainshaft rear bearing using the SSTs.



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2W/4W Clutch Hub Disassembly Note

1. Remove the 2W/4W clutch hub using the SSTs.

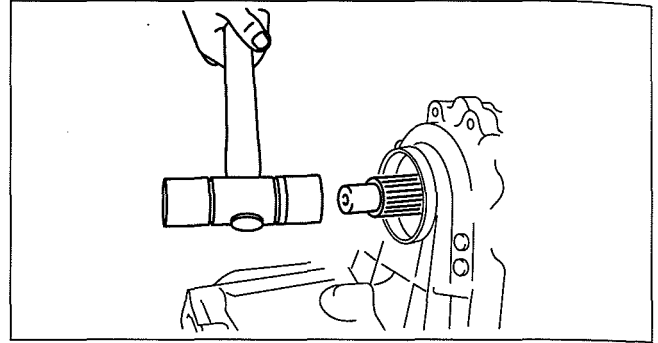


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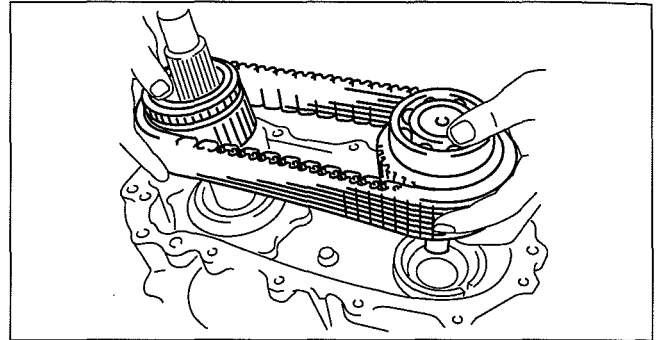
Chain, Front Drive Shaft and Sprocket Disassembly Note

1. Remove the front drive shaft using the plastic hammer.



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2. Remove the chain, front drive shaft and sprocket as a single unit.

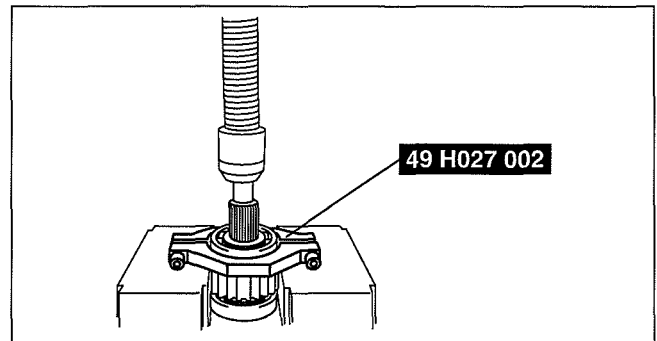


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Front Drive Shaft Bearing Disassembly Note

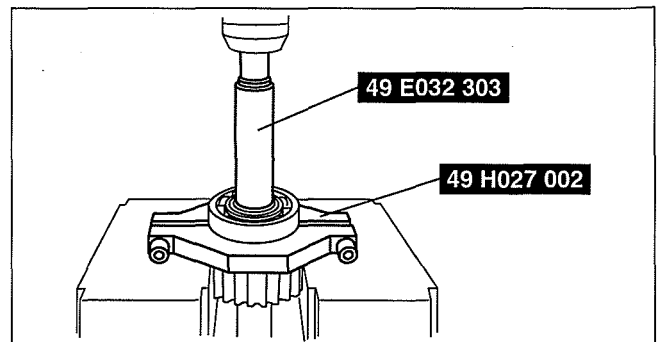
1. Remove the front drive shaft bearing using the SSTs and press.

Center case side



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Rear case side

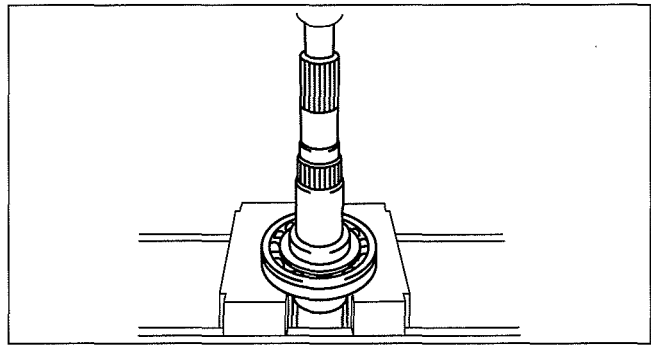


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Mainshaft Center Bearing Disassembly Note

1. Remove the mainshaft front bearing using the press.

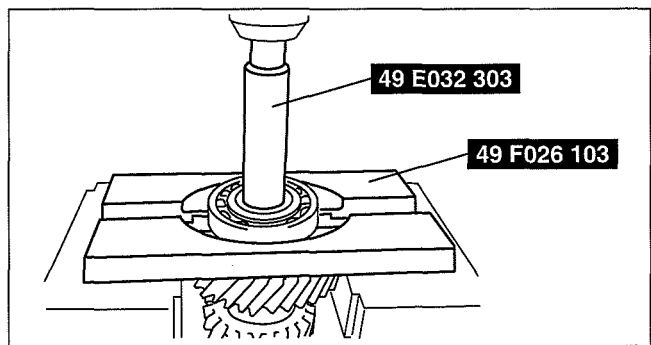


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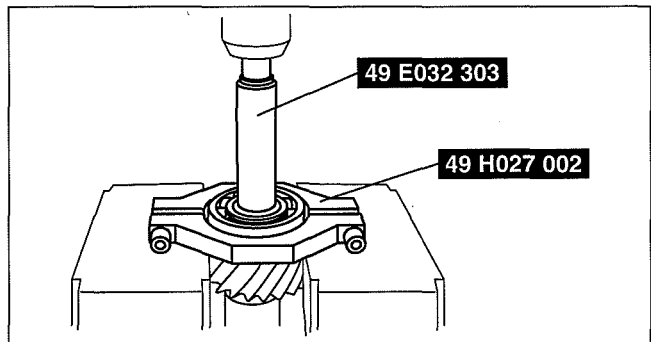
Countershaft Bearing Disassembly Note

1. Remove the countershaft bearing using the SSTs and press.
- Front case side



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Center case side



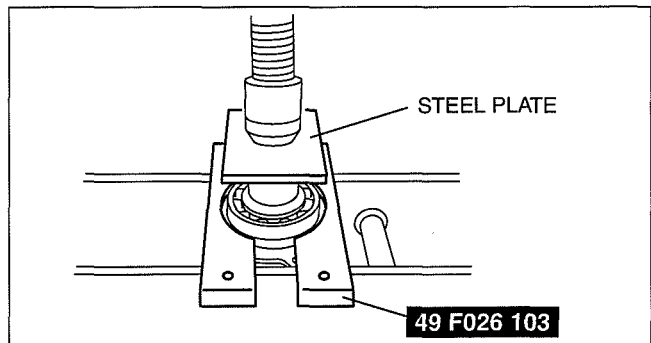
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Maindrive Gear Bearing Disassembly Note

1. Remove the maindrive gear bearing using the SST, press and steel plate.

Note

- Steel plate height: 10 mm {0.39 in} or more.

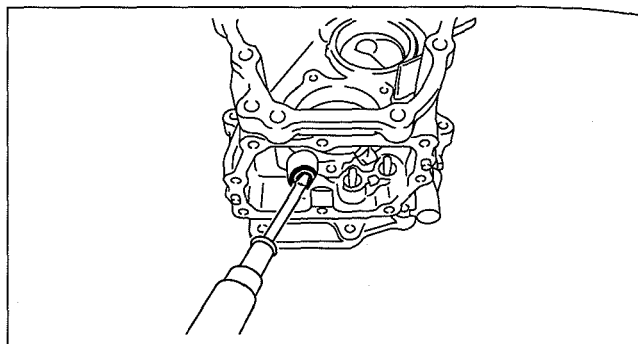


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TRANSFER

Oil Seal (Control Rod) Disassembly Note

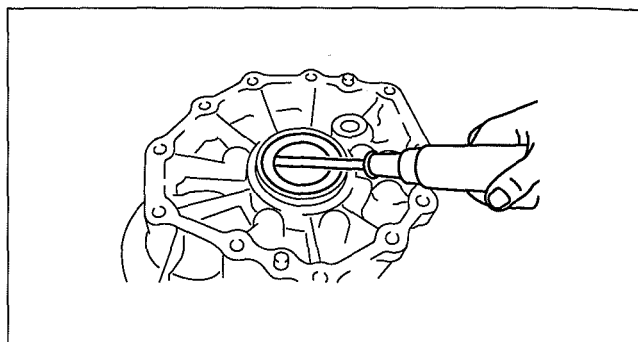
1. Remove the oil seal (control rod) using a flathead screwdriver as shown in the figure.



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Oil Seal (Front Case) Disassembly Note

1. Remove the oil seal (front case) using a flathead screwdriver as shown in the figure.



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TRANSFER ASSEMBLY

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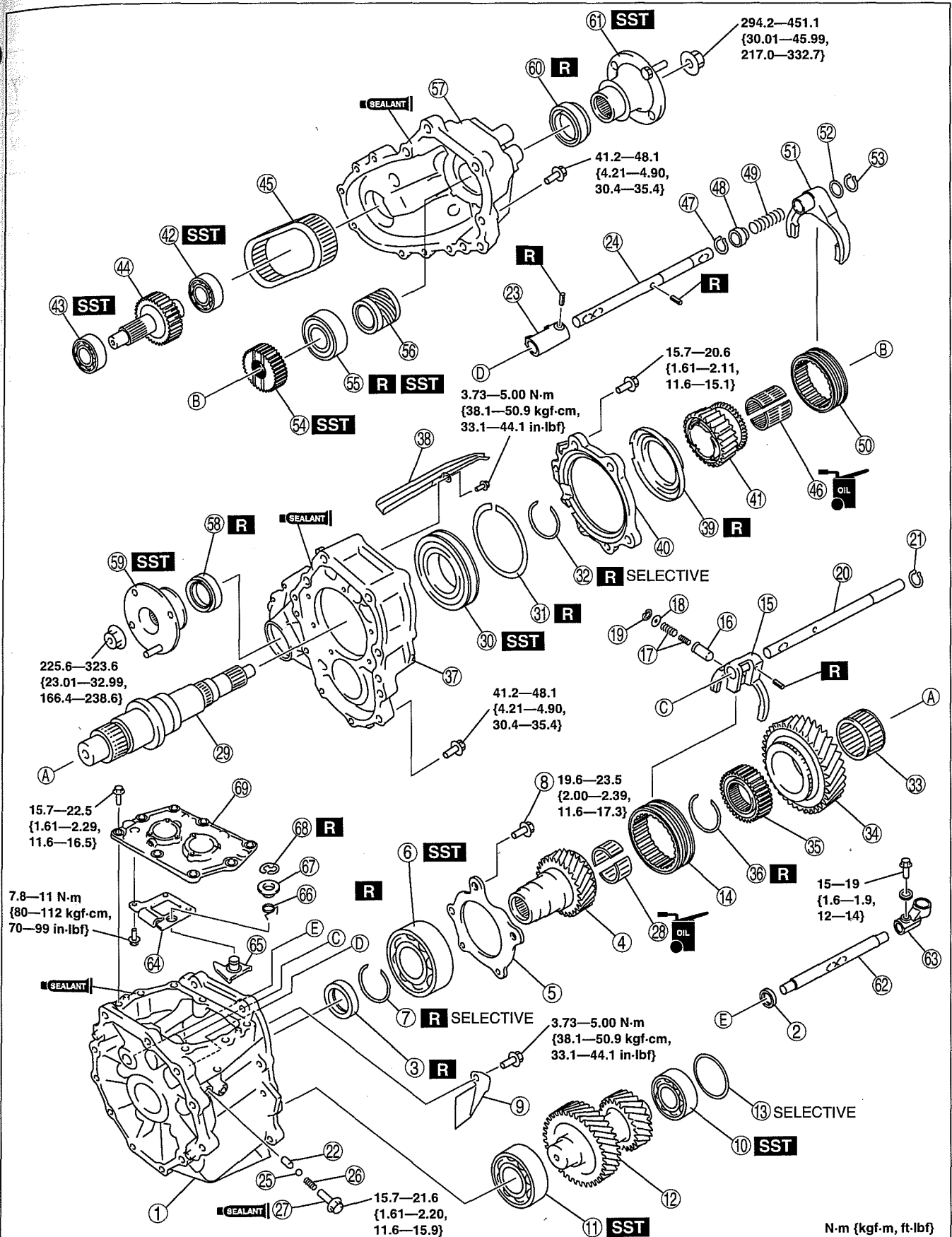
Before Service Precautions

- Assemble with bare hands or using vinyl gloves. To prevent foreign material from entering the transfer, do not use cotton work gloves or a rag.
- Apply sufficient gear oil to the sliding surfaces and O-rings, and be careful not to damage when assembling.
- Replace the transfer with a new one if the case alignment surface is damaged. Be careful not to damage it since it may cause oil leakage.
- When installing silicone sealant, clean off the old sealant adhering to the sealing area and clean the sealing area with cleaning fluids.
- After installing a seal, leave the parts alone for **2 h or more**. Do not add oil or operate the vehicle during this time.

TRANSFER

Transfer Assembly

03



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1 Front case

2 Oil seal (control rod)
(See 03-16-11 Oil Seal (control rod) Assembly Note.)

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3	Oil seal (front case) (See 03-16-11 Oil Seal (front case) Assembly Note.)
4	Maindrive gear
5	Retainer
6	Maindrive gear bearing (See 03-16-11 Maindrive Gear Bearing Assembly Note.)
7	Retaining ring (See 03-16-11 Maindrive Gear Bearing Assembly Note.)
8	Retaining bolt
9	Baffle plate
10	Countershaft rear bearing (center case side) (See 03-16-12 Countershaft Bearing Assembly Note.)
11	Countershaft front bearing (front case side) (See 03-16-12 Countershaft Bearing Assembly Note.)
12	Countershaft (See 03-16-12 Countershaft Assembly Note.)
13	Adjustment shim (See 03-16-12 Countershaft Assembly Note.)
14	H/L hub sleeve
15	H/L shift fork
16	Plunger
17	Spring
18	Plain washer
19	Snap ring
20	H/L shift rod
21	Snap ring
22	Interlock pin
23	2W/4W shift end
24	2W/4W shift rod
25	Steel ball
26	Spring
27	Retaining bolt
28	Needle bearing
29	Mainshaft (See 03-16-13 Mainshaft and Mainshaft Center Bearing Assembly Note.)
30	Mainshaft center bearing (See 03-16-13 Mainshaft and Mainshaft Center Bearing Assembly Note.)
31	Retaining ring
32	Retaining ring (See 03-16-13 Mainshaft and Mainshaft Center Bearing Assembly Note.)
33	Needle bearing
34	Low gear
35	H/L clutch hub
36	Retaining ring
37	Center case (See 03-16-13 Center Case Assembly Note.)
38	Oil passage (See 03-16-14 Oil Passage Assembly Note.)
39	Oil catcher (See 03-16-14 Oil Catcher Assembly Note.)
40	Center bearing retainer

41	Sprocket (See 03-16-14 Chain, Front Drive Shaft and Sprocket Assembly Note.)
42	Front drive shaft bearing (rear case side) (See 03-16-15 Front Drive Shaft Bearing Assembly Note.)
43	Front drive shaft bearing (center case side) (See 03-16-15 Front Drive Shaft Bearing Assembly Note.)
44	Front drive shaft (See 03-16-14 Chain, Front Drive Shaft and Sprocket Assembly Note.)
45	Chain (See 03-16-14 Chain, Front Drive Shaft and Sprocket Assembly Note.)
46	Needle bearing (See 03-16-14 Chain, Front Drive Shaft and Sprocket Assembly Note.)
47	Snap ring
48	Collar
49	Spring
50	2W/4W hub sleeve
51	2W/4W shift fork
52	Spacer
53	Snap ring
54	2W/4W clutch hub (See 03-16-15 2W/4W Clutch Hub Assembly Note.)
55	Mainshaft rear bearing (See 03-16-16 Mainshaft Rear Bearing Assembly Note.)
56	Speed drive gear
57	Rear case (See 03-16-16 Rear Case Assembly Note.)
58	Oil seal (center case) (See 03-16-16 Oil Seal (center case) Assembly Note.)
59	Companion flange (center case side)
60	Oil seal (rear case) (See 03-16-17 Oil Seal (rear case) Assembly Note.)
61	Companion flange (rear case side)
62	Control rod
63	Control rod end
64	Guide plate
65	Reverse gate lever
66	Reverse gate spring
67	Spacer
68	E-ring
69	Control case (See 03-16-17 Control Case Assembly Note.)

TRANSFER

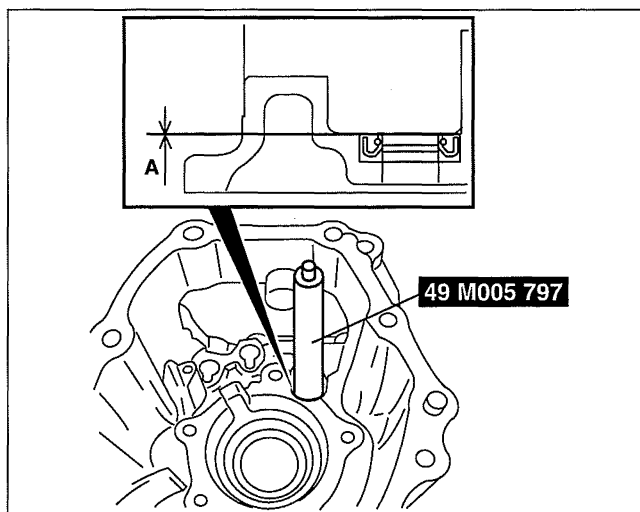
Oil Seal (control rod) Assembly Note

Caution

- Assemble the oil seal being careful not to assemble it in the incorrect direction.

1. Apply specified grease to the lip of a new oil seal.
2. Install the oil seal (control rod) using the SST.

Installation depth A: 0 mm {0 in}

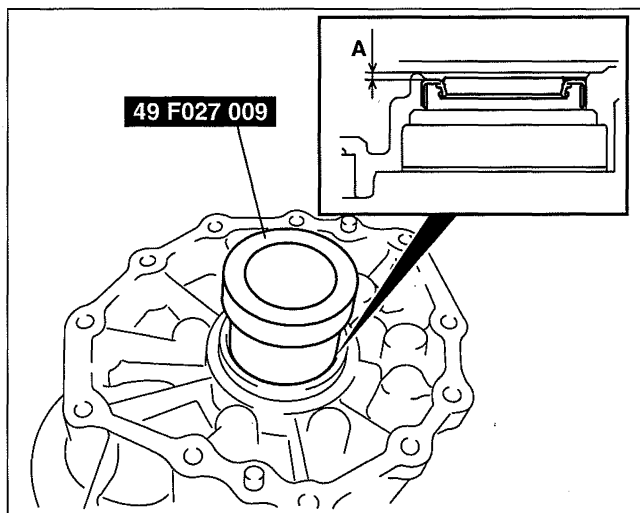


DBG316BMB030

Oil Seal (front case) Assembly Note

1. Apply specified grease to the lip of a new oil seal.
2. Install the oil seal (front case) using the SST.

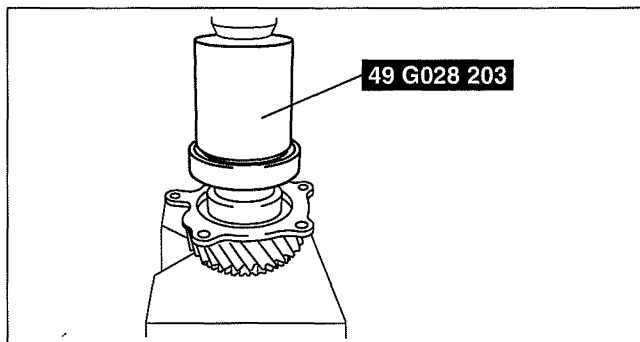
Installation depth A: 2.4—3.0 mm {0.095—0.118 in}



DBG316BMB013

Maindrive Gear Bearing Assembly Note

1. Assemble the maindrive gear bearing using the SST and press.
2. Install the retaining ring.



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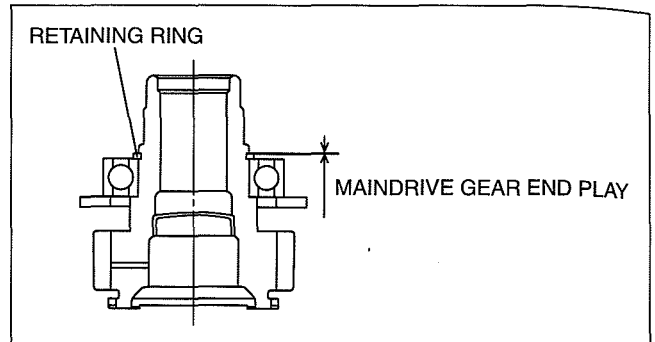
TRANSFER

3. Measure the clearance between the retaining ring and groove of the maindrive gear.
 - If not within the specification, adjust by choosing the proper retaining ring.

Maindrive gear bearing end play
 0—0.15 mm {0.0—0.0059 in}

Maindrive gear bearing retaining ring

Thickness (mm {in})
2.60 {0.102}
2.67 {0.105}
2.74 {0.108}

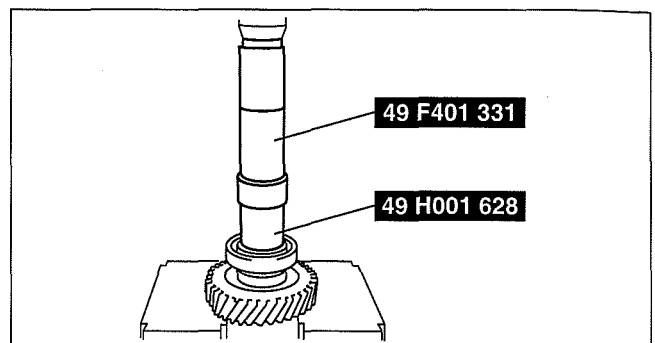


DBG316BMB033

Countershaft Bearing Assembly Note

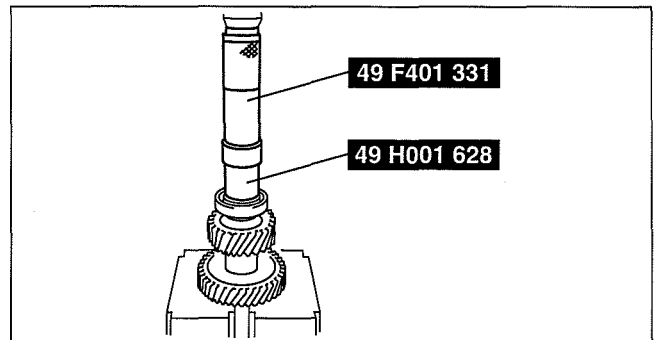
1. Assemble the countershaft bearing using the SSTs and press.

Front case side



DBG316BMB019

Center case side



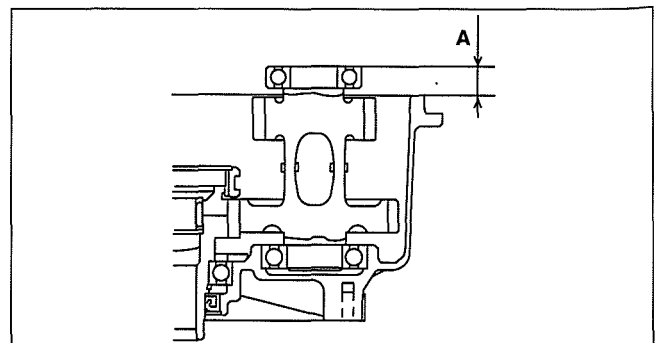
DBG316BMB020

Countershaft Assembly Note

1. Assemble the countershaft to the front case.
2. Measure the depth A as shown in the figure.
3. Select the correct countershaft adjustment shim thickness.

Countershaft adjustment shim selective chart

Dimension A (mm {in})	Shim thickness (mm {in})
25.0—25.1 {0.985—0.988}	0.5 {0.020}
25.1—25.2 {0.989—0.992}	0.4 {0.016}
25.2—25.3 {0.993—0.996}	0.3 {0.012}
25.3—25.4 {0.997—1.000}	0.2 {0.008}
25.4—25.5 {1.000—1.003}	0.1 {0.004}
25.5—25.6 {1.004—1.007}	-



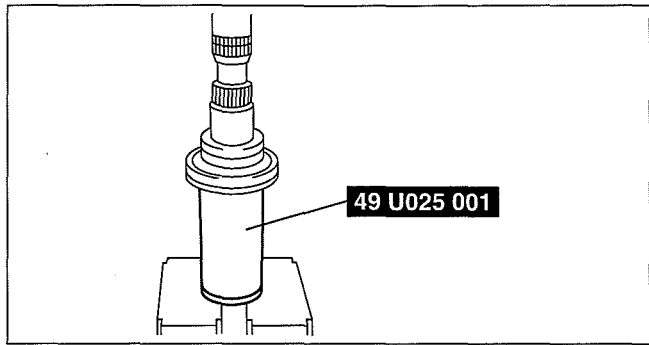
DBG316BMB032

4. Install the adjustment shim.

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Mainshaft and Mainshaft Center Bearing Assembly Note

1. Assemble the mainshaft center bearing using the SST and press.

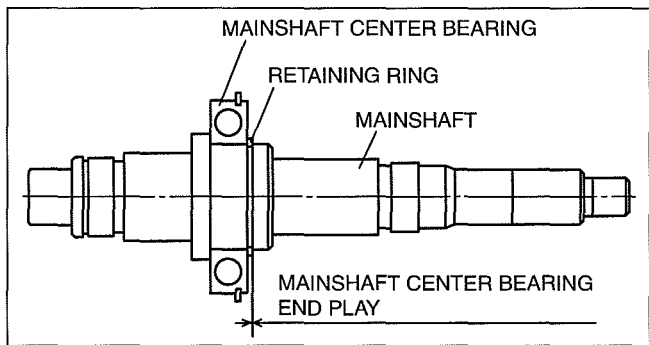


2. Measure the clearance between the retaining ring and groove of the mainshaft.
 - If not within the specification, adjust by choosing the proper retaining ring.

Mainshaft center bearing end play
0—0.15 mm {0.0—0.0059 in}

Mainshaft center bearing retaining ring

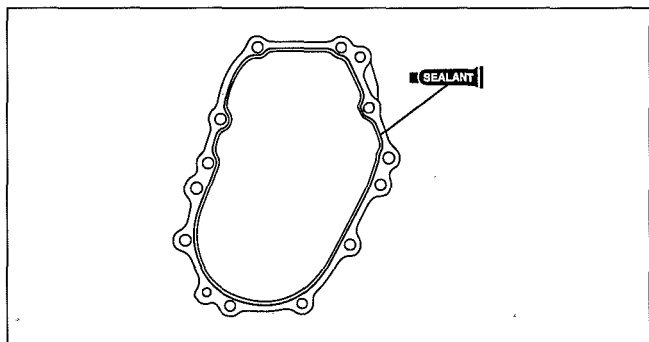
Thickness (mm {in})
3.1 {0.122}
3.2 {0.126}
3.3 {0.130}



3. Assemble the mainshaft to the center case using the plastic hammer.

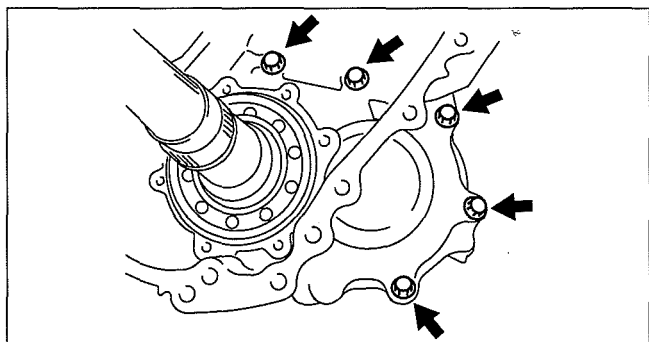
Center Case Assembly Note

1. Apply sealant to the contact surfaces of the front case and center case as shown in the figure.



2. Assemble the center case.

Bolt length (measured from below the head)
43 mm {1.69 in}



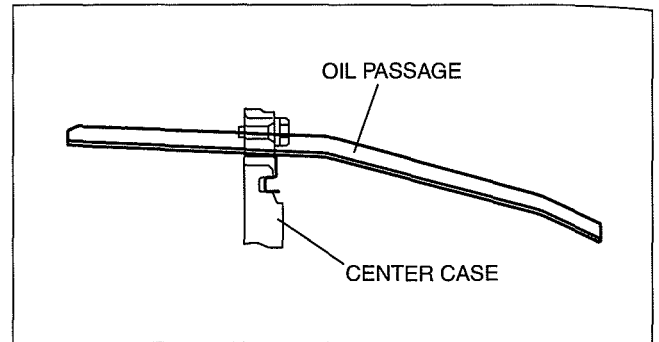
TRANSFER

Oil Passage Assembly Note

1. Assemble the oil passage to the center case.

Note

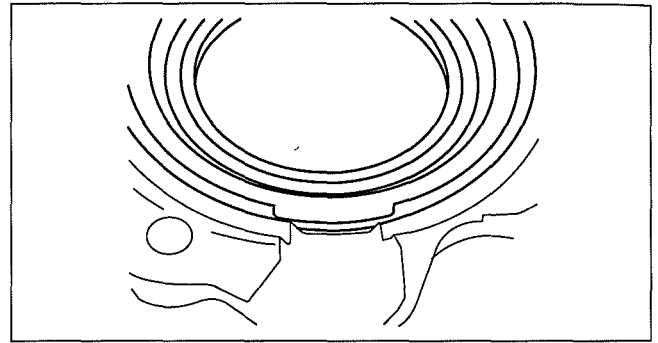
- Align the oil passage of the center case groove as shown in the figure.



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Oil Catcher Assembly Note

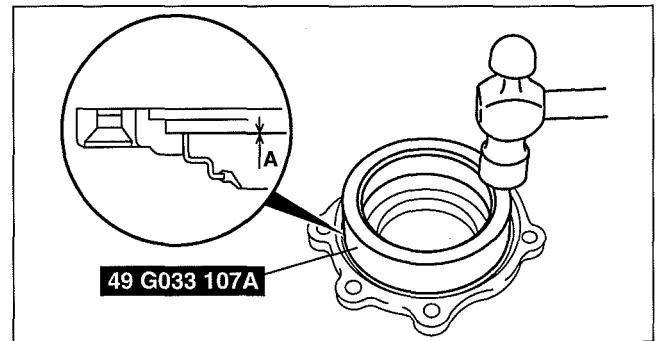
1. Align the oil catcher groove as shown in the figure.
2. Apply specified grease to the lip of a new oil seal.



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3. Install the oil catcher using the SST.

Installation depth A: 0 mm {0 in}



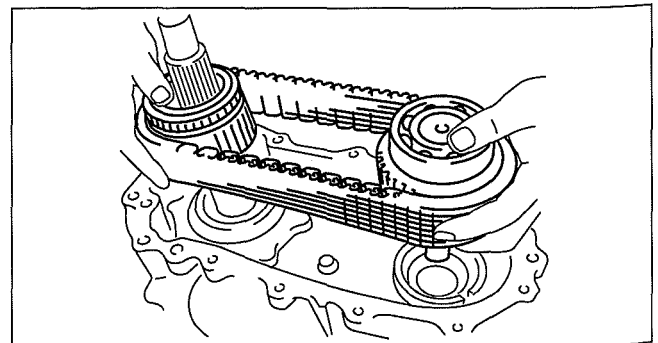
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Chain, Front Drive Shaft and Sprocket Assembly Note

1. Assemble the chain, front drive shaft and sprocket as a single unit.

Note

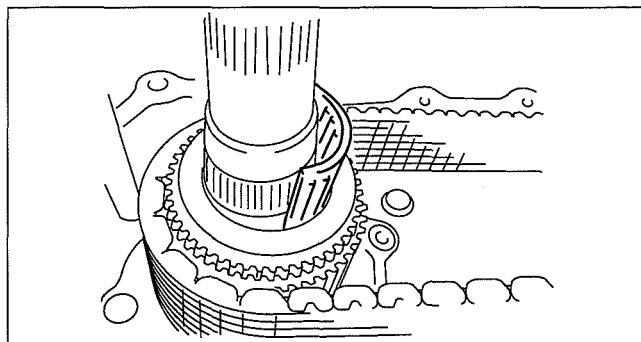
- Assemble the front drive shaft using the plastic hammer.



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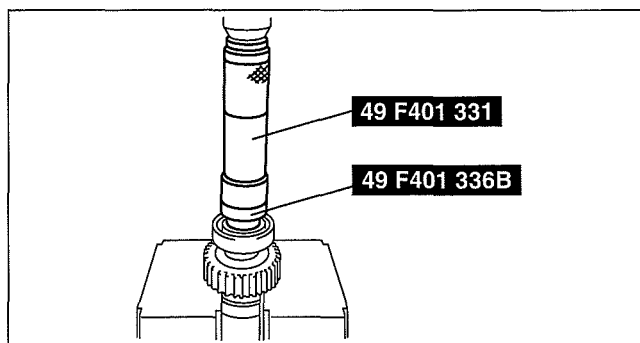
2. Install the needle bearings while turning the mainshaft.



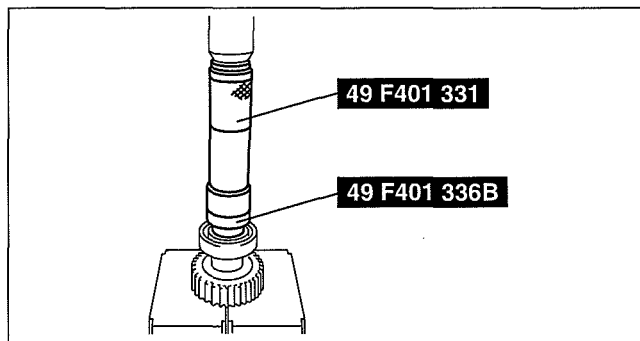
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Front Drive Shaft Bearing Assembly Note

1. Assemble the front drive shaft bearing using the SSTs and press.
Center case side

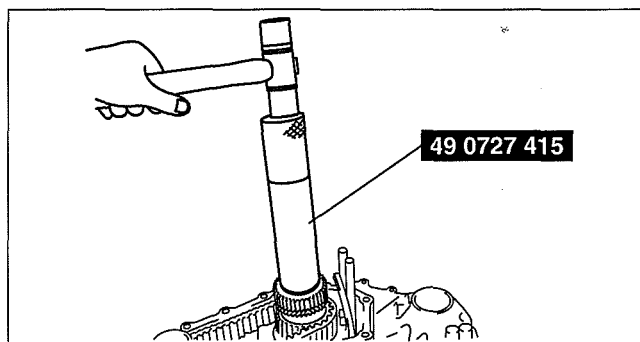


Rear case side



2W/4W Clutch Hub Assembly Note

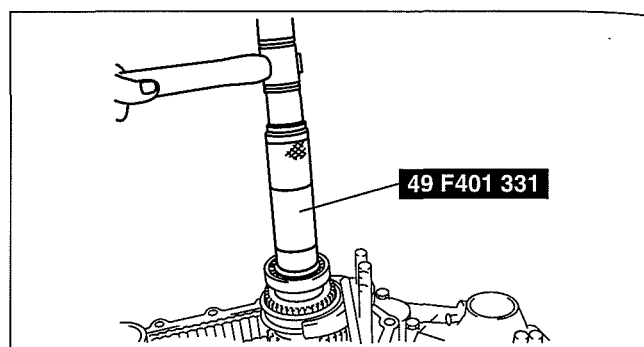
1. Assemble the 2W/4W clutch hub using the SST.



TRANSFER

Mainshaft Rear Bearing Assembly Note

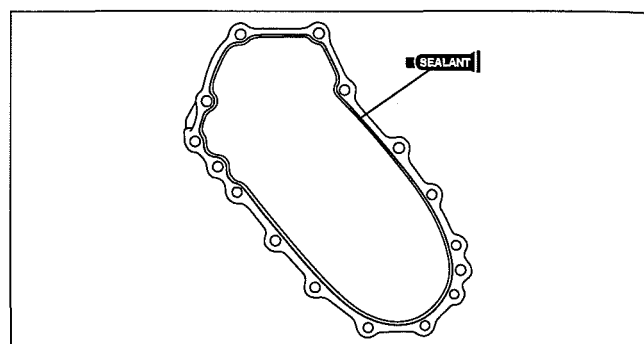
1. Assemble the mainshaft rear bearing using the SST.



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Rear Case Assembly Note

1. Apply sealant to the contact surfaces of the center case and rear case as shown in the figure.



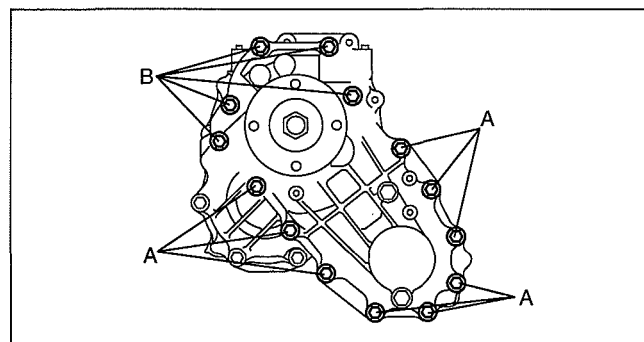
DBG316BMB029

2. Assemble the rear case.

Bolt length (measured from below the head)

A: 58 mm {2.28 in}

B: 113 mm {4.45 in}

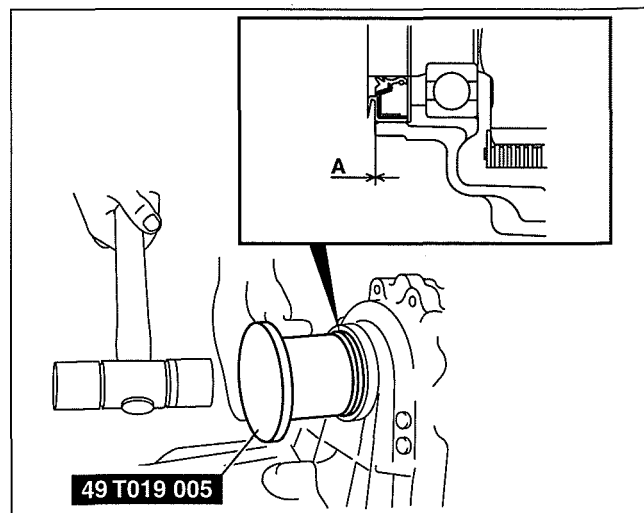


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Oil Seal (center case) Assembly Note

1. Apply specified grease to the lip of a new oil seal.
2. Install the oil seal (center case) using the SST.

Installation depth A: 0 mm {0 in}



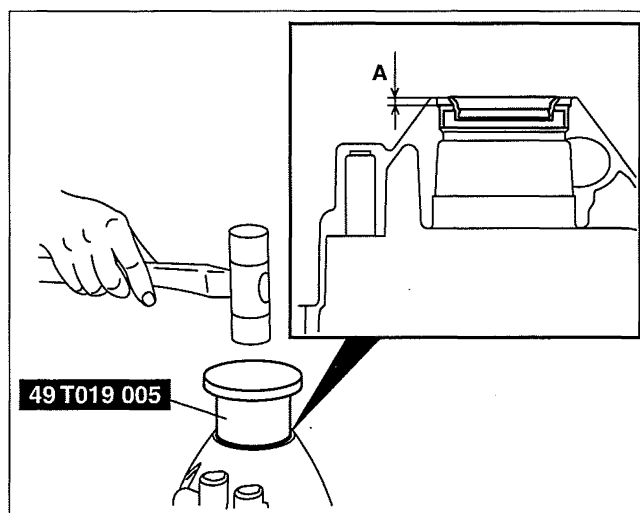
DBG316BMB014

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Oil Seal (rear case) Assembly Note

1. Apply specified grease to the lip of a new oil seal.
2. Install the oil seal (rear case) using the **SST**.

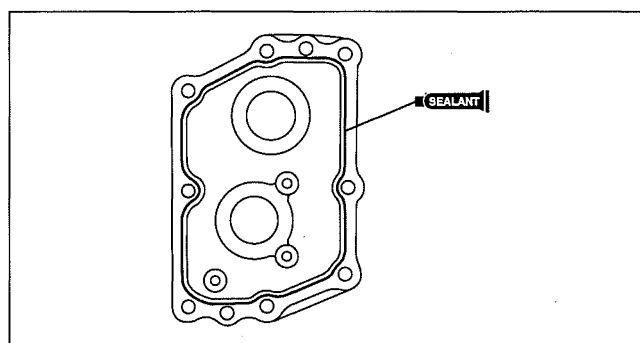
Installation depth A: 6.7—7.3 mm {0.27—0.28 in}



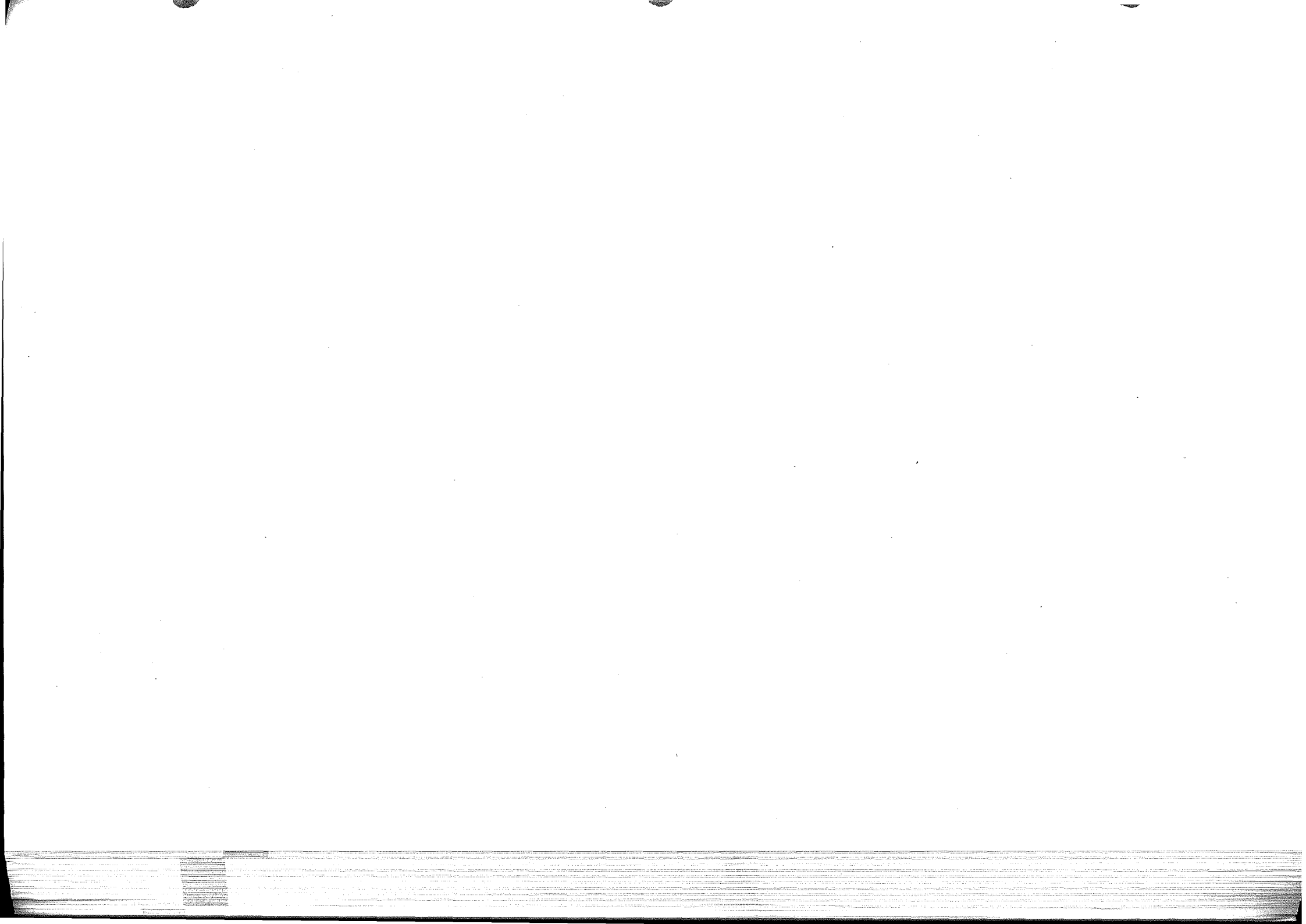
DBG316BMB015

Control Case Assembly Note

1. Apply sealant to the contact surfaces of the control case and transfer case as shown in the figure.
2. Install the control case to the transfer case.



DBG511BMB021



03-16 TRANSFER [5R55S]

TRANSFER CASE CLEANING

[5R55S] 03-16-1

TRANSFER CASE DISASSEMBLY

[5R55S] 03-16-2

TRANSFER CASE ASSEMBLY

[5R55S] 03-16-14

TRANSFER CASE INSPECTION

[5R55S] 03-16-26

TRANSFER CASE CLEANING[5R55S]

id0316c1501000

03

Note

- Before cleaning, check the magnet for the presence of metal particles which indicate internal chipping of the transfer case.

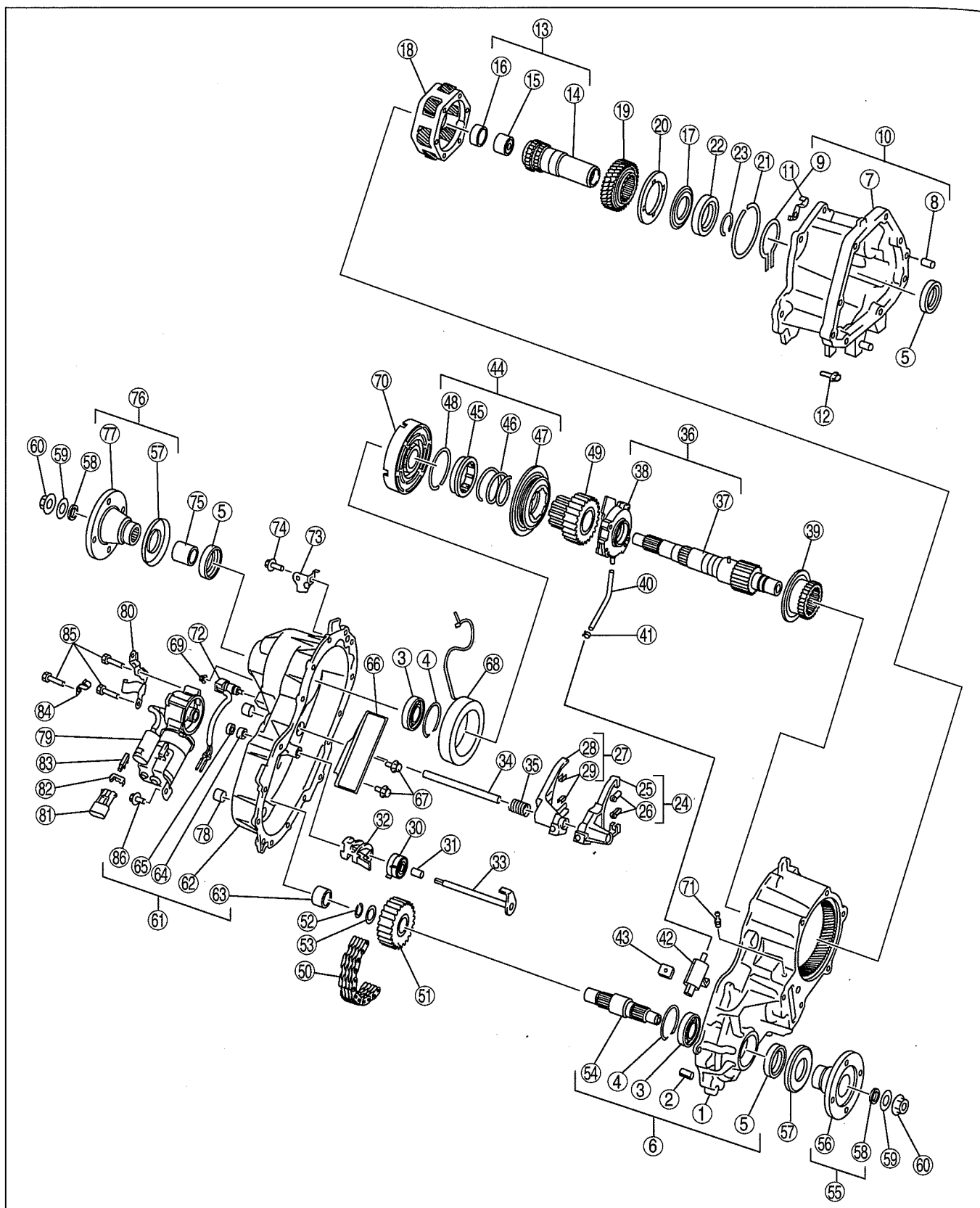
1. Using a cleaning solvent, clean the old oil and dirt deposits.
2. After cleaning dry the parts with low pressure (138 kPa {1.41 kgf/cm², 20 psi maximum}) compressed air.
3. Lubricate the ball bearings and needle bearing with oil. Protect lubricated bearings from dust.

TRANSFER [5R55S]

TRANSFER CASE DISASSEMBLY[5R55S]

id0316c1501100

Exploded View



b5f5za0000446

1	Center transfer case
2	Dowel pin
3	Bearing
4	Snap ring

5	Oil seal
6	Center transfer case component
7	Front transfer case
8	Dowel pin

TRANSFER [5R55S]

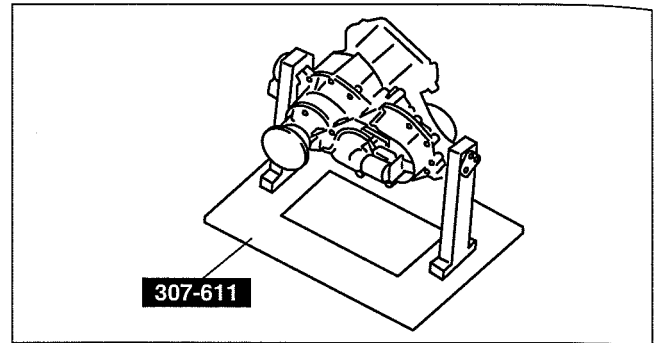
9	Snap ring
10	Front transfer case component
11	Clip
12	Bolt
13	Input shaft component
14	Input shaft
15	Needle bearing
16	Bushing
17	Thrust washer
18	Planetary gear component
19	Sun gear
20	Thrust plate
21	Snap ring A
22	Ball bearing
23	Snap ring B
24	Reduction shift fork component
25	Reduction shift fork
26	Reduction shift fork facing
27	Lockup shift fork component
28	Lockup shift fork
29	Lockup shift fork facing
30	Spring
31	Spacer
32	Cam
33	Shift shaft
34	Rail shift
35	Return spring
36	Output shaft and gerotor pump component
37	Output shaft component
38	Gerotor pump component
39	Reduction hub
40	Pump hose
41	Hose clamp
42	Oil strainer
43	Magnet
44	Lockup component
45	Lockup hub
46	Spring
47	Lockup collar

48	Snap ring
49	Drive sprocket
50	Drive chain
51	Driven sprocket
52	Snap ring
53	Spacer
54	Lower output shaft
55	Front output flange component
56	Front output flange
57	Deflector
58	Oil seal
59	Washer
60	Locknut
61	Rear transfer case component
62	Rear transfer case
63	Needle bearing
64	Sleeve
65	Seal
66	Snubber
67	Bolt
68	Clutch coil
69	Nut
70	Clutch housing
71	Breather pipe
72	Speed sensor component
73	Bracket
74	Bolt
75	Spacer
76	Rear companion flange component
77	Rear companion flange
78	Oil plug
79	Motor component
80	Bracket
81	Terminal connector
82	Clip
83	Bracket
84	Clip
85	Bolt
86	Bolt

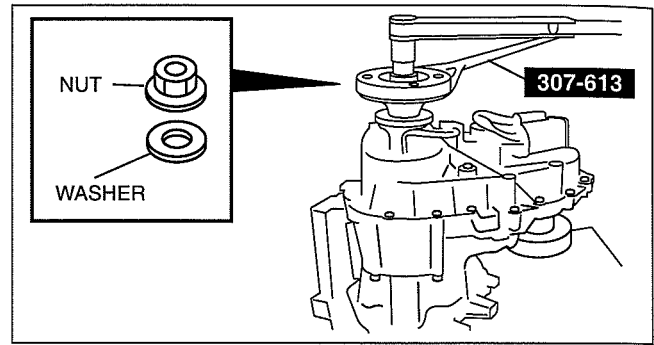
TRANSFER [5R55S]

Disassembly Procedure

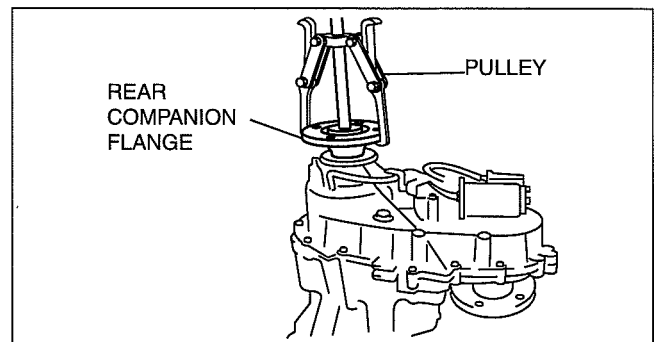
1. Using the **SST**, set the transfer case as shown in the figure.



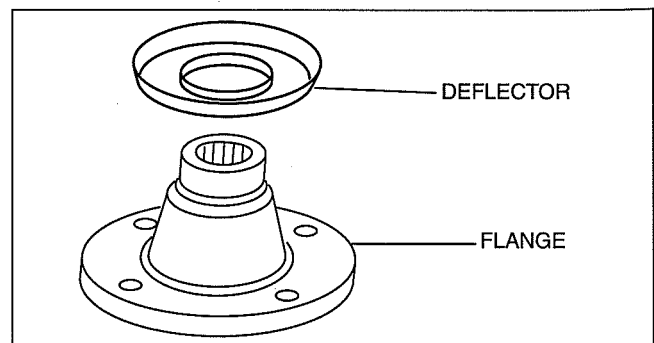
2. Secure the rear companion flange component using the **SST**.
3. Remove the nut and washer.



4. Remove the rear companion flange using the pulley.

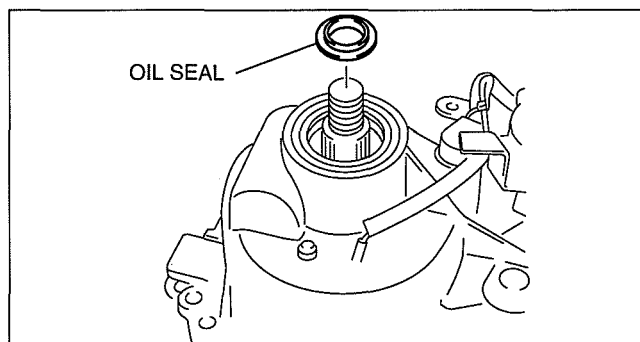


5. Remove the deflector from the flange.



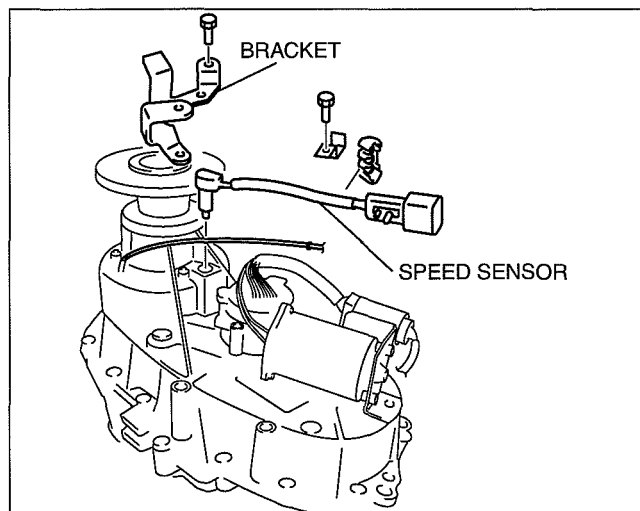
TRANSFER [5R55S]

6. Remove the oil seal from the output shaft.



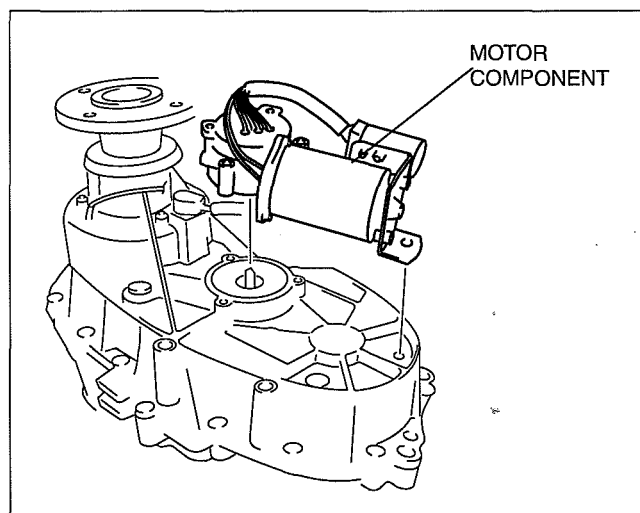
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7. Remove the bracket and speed sensor.



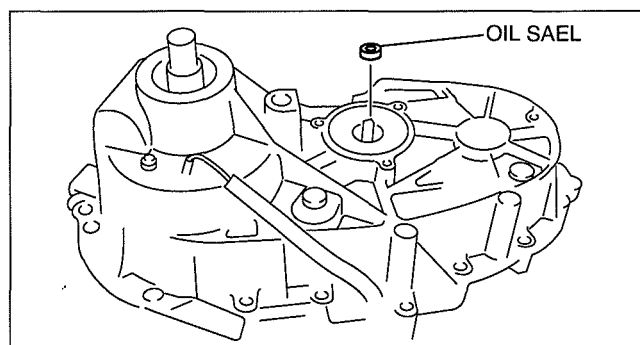
arnffv00000614

8. Remove the motor component.



arnffv00000619

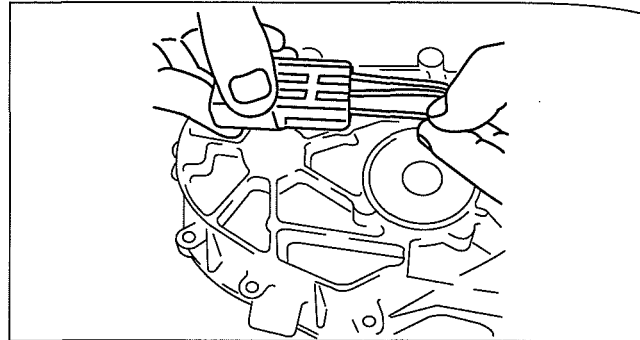
9. Remove the oil seal.



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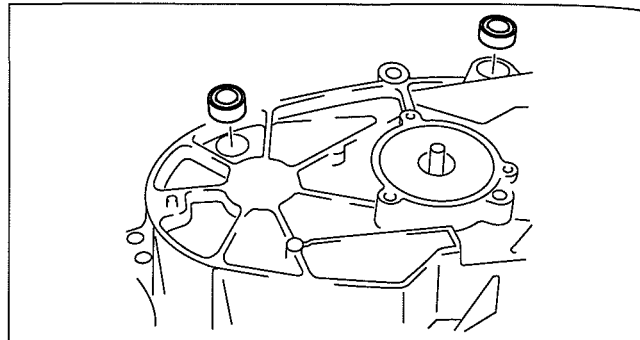
TRANSFER [5R55S]

10. Remove the clutch coil terminal from the connector and pull out the wiring harness from the sleeve.



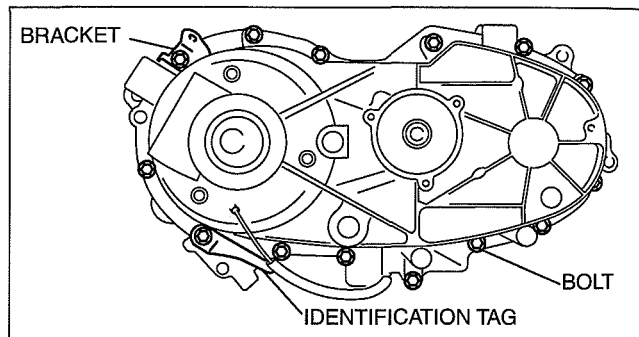
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11. Remove the two oil plugs from the rear transfer case.



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12. Remove the 14 bolts and the identification tag.

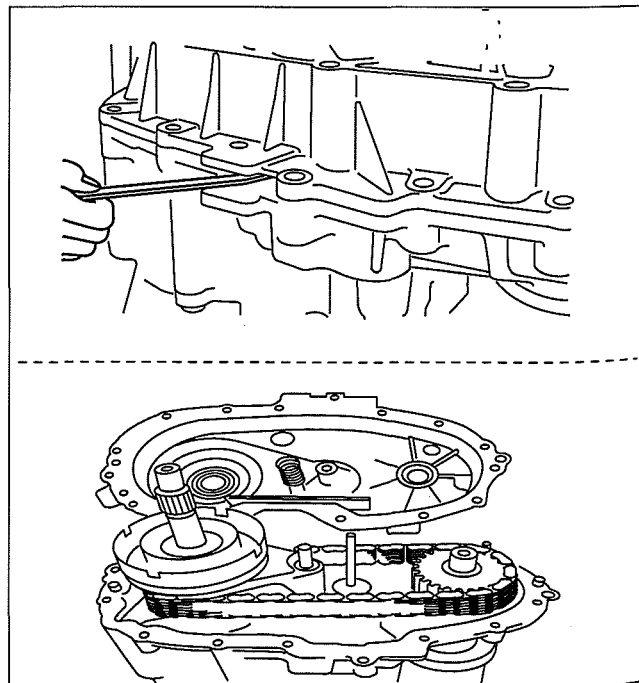


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13. Pry at the transfer case projection as shown in the figure a flathead screwdriver, and remove the rear transfer case from the center transfer case.

Caution

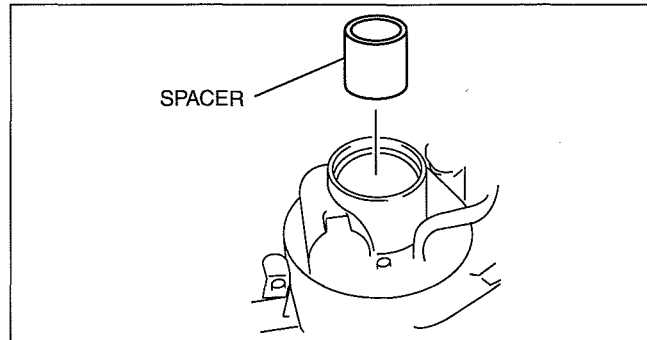
- When prying at the transfer case, be careful not to damage the case.



b5r5za00000411

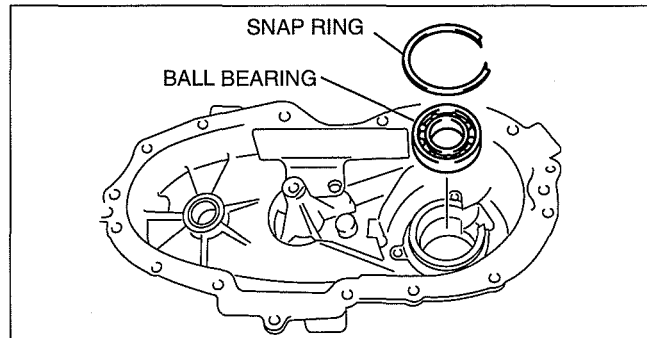
TRANSFER [5R55S]

14. Remove the spacer.
15. Using a flathead screwdriver, remove the oil seal.



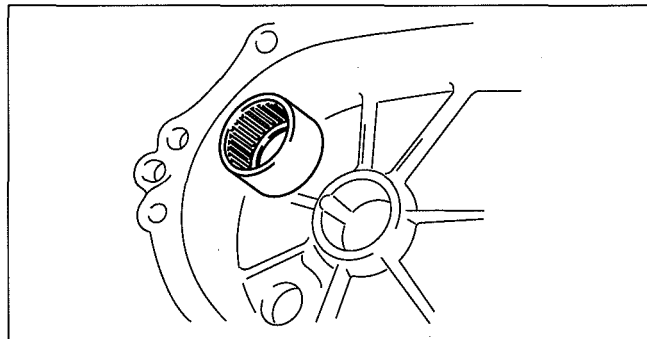
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16. Remove the snap ring.
17. Pull out the ball bearing from the rear transfer case.



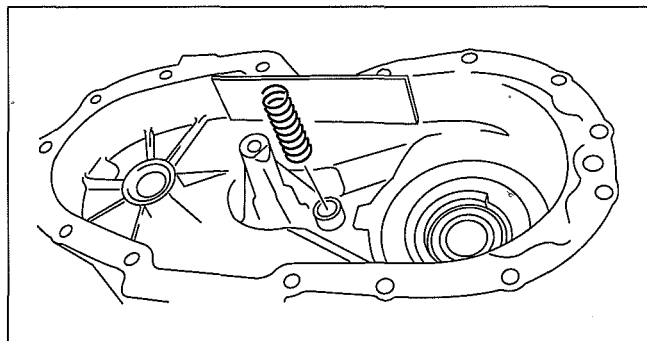
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18. Pull out the needle bearing from the rear transfer case.



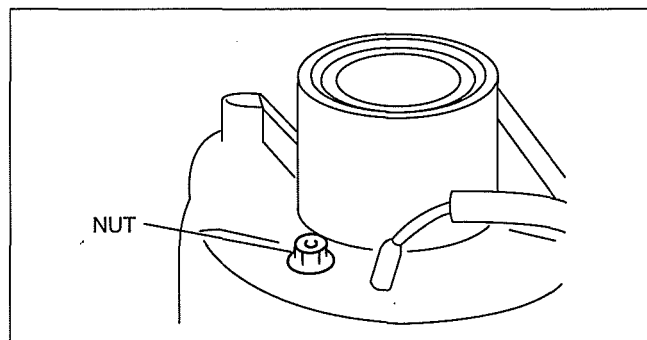
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19. Remove the return spring.



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20. Remove the 3 nuts.
21. Remove the clutch coil from the rear transfer case.



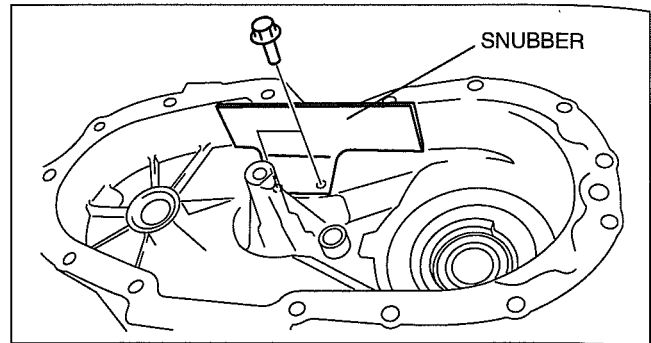
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TRANSFER [5R55S]

22. Remove the snubber from the rear transfer case.
23. Clean and remove the sealant of the rear transfer case and center transfer case.

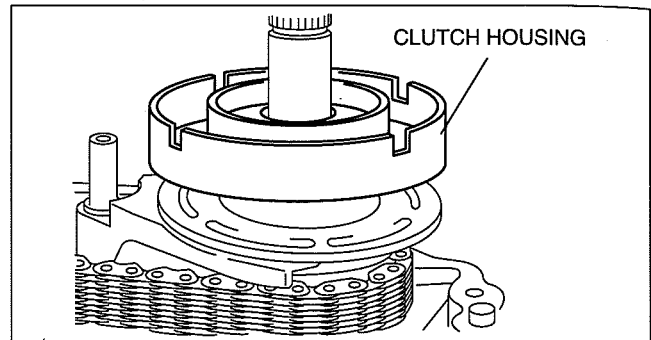
Caution

- Be careful not to damage the metal surface.



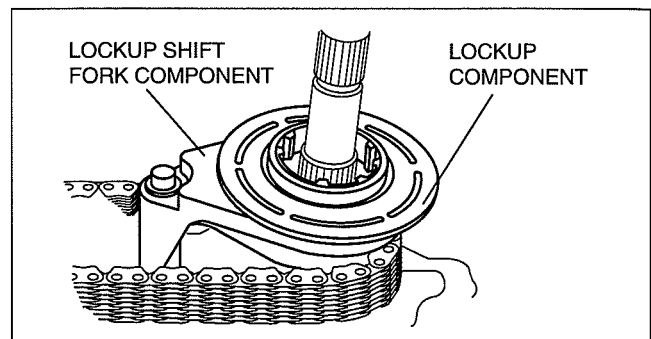
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24. Remove the clutch housing from the output shaft.



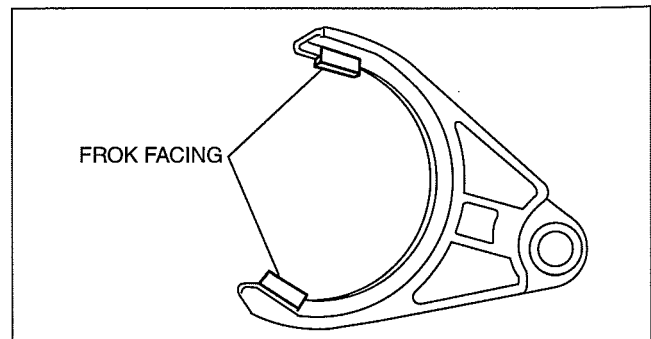
b5r5za00000312

25. Slide the lockup component and the lockup shift fork component as a single unit to remove the lockup component from the output shaft.
26. Remove the lockup shift fork component from the lockup component.



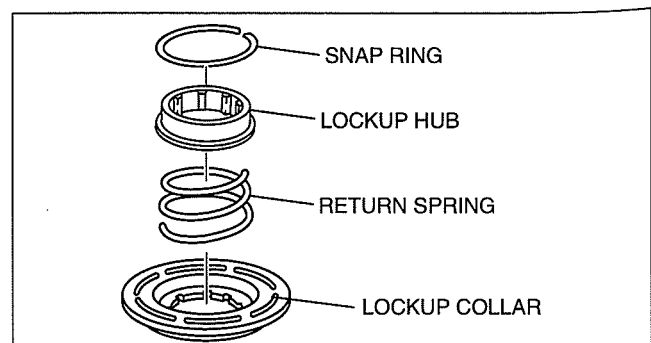
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27. Remove the two fork facings from the lockup shift fork.



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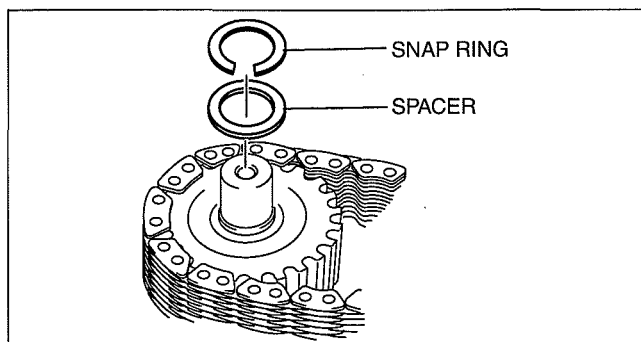
28. Remove the snap ring, lockup hub and return spring from the lockup collar.



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TRANSFER [5R55S]

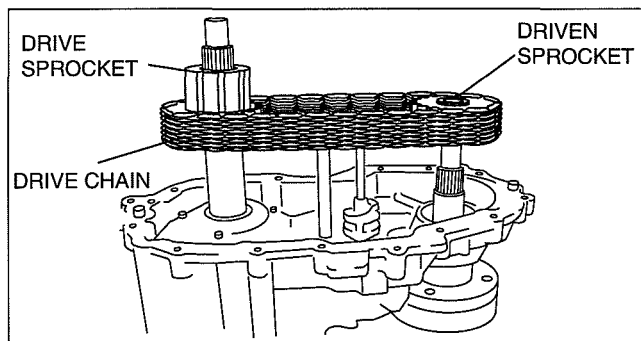
29. Remove the snap ring and spacer from the output shaft.



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30. Remove the drive chain, driven sprocket and drive sprocket from the output shaft at a time.

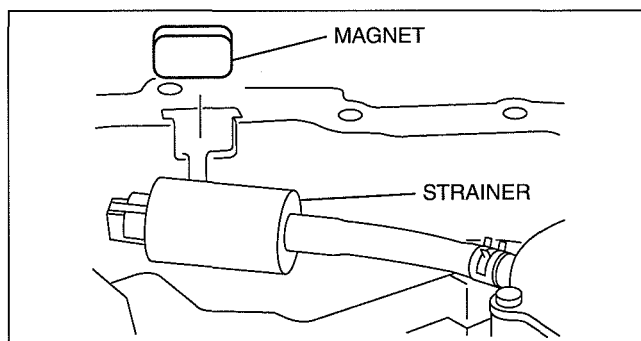
31. Separate the drive chain and sprockets.



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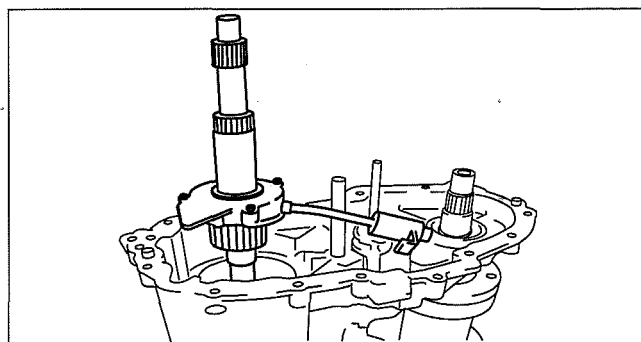
32. Remove the magnet from the slot in the center transfer case.

33. Remove the strainer from the center transfer case.



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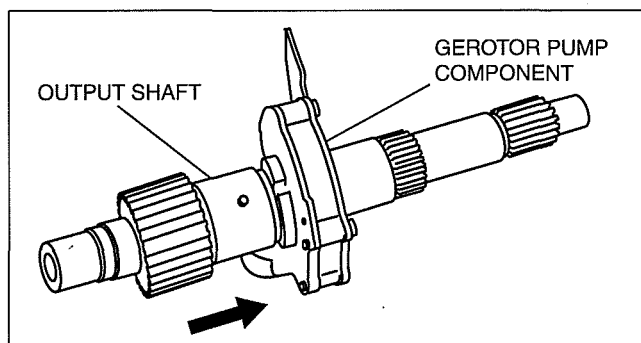
34. Remove the output shaft component and gerotor pump component from the center transfer case.



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35. Remove the gerotor pump component from the output shaft.

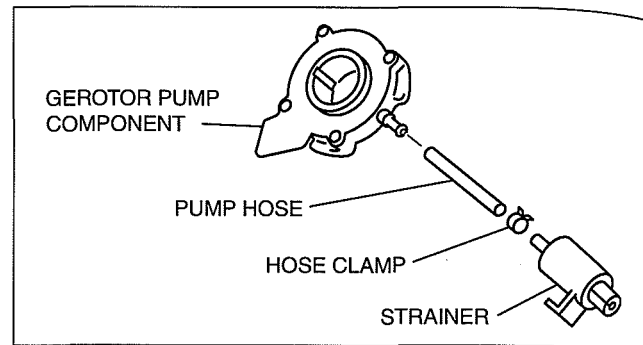
36. Loosen the hose clamp and remove the pump hose from the gerotor pump component.



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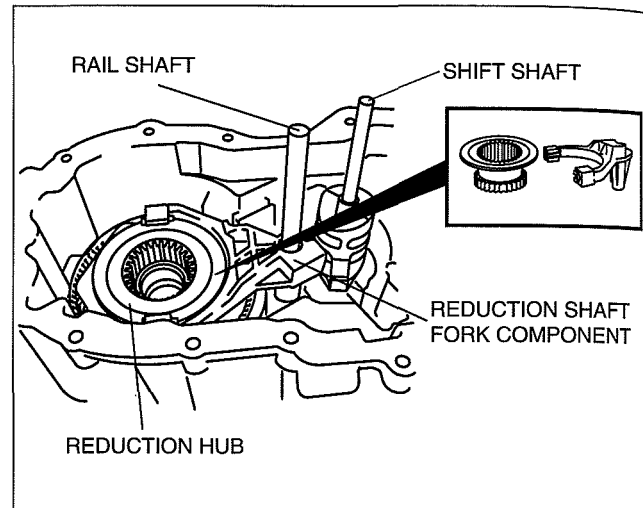
TRANSFER [5R55S]

37. Remove the hose clamp, pump hose and strainer.



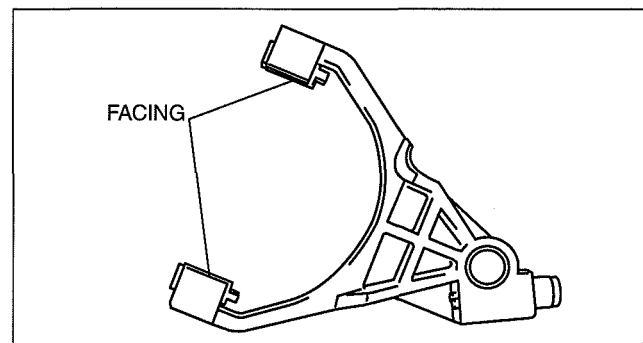
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38. Remove the reduction hub, reduction shift fork component, rail shaft and shift shaft component from the center transfer case.



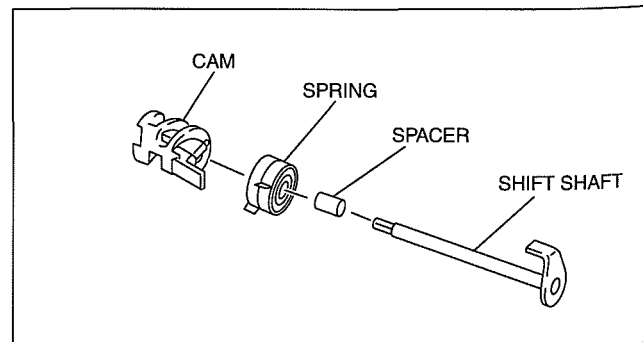
b5r5za00000320

39. Remove the two shift fork facings from the reduction shift fork component.



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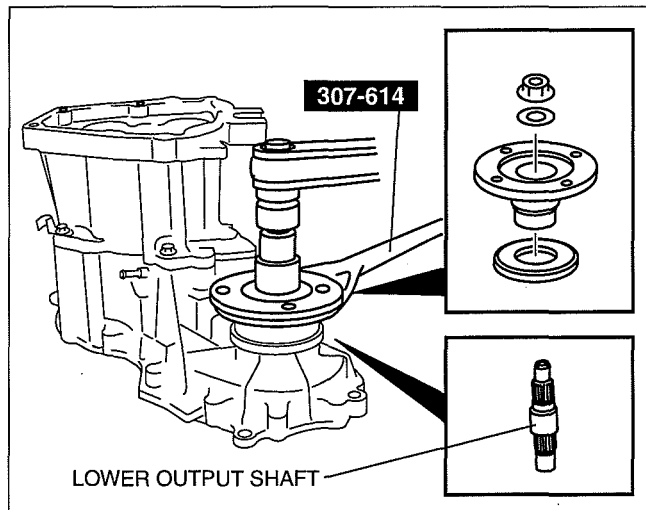
40. Disassemble the cam, spring, spacer, and shift shaft.



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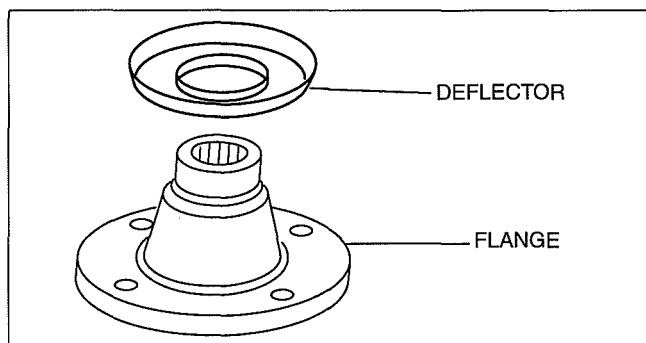
TRANSFER [5R55S]

41. Secure the flange using the SST.
42. Remove the nut and washer.



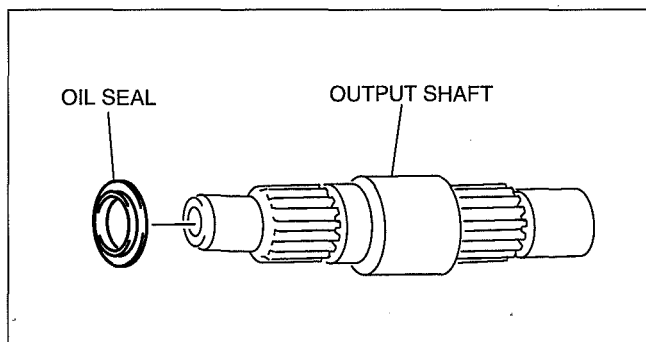
amftv00000555

43. Remove the deflector from the flange.
44. Pull out the front output flange component.
45. Remove the lower output shaft.



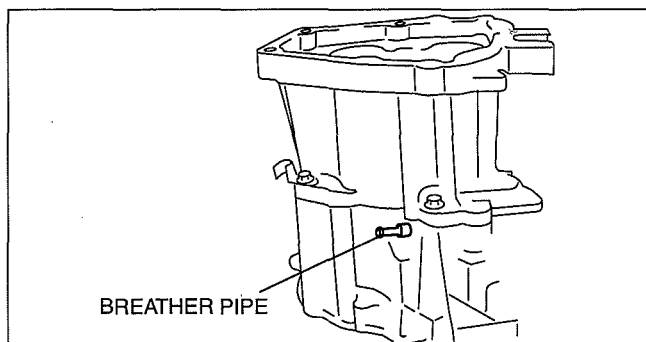
amftv00000616

46. Remove the oil seal from the lower output shaft.
47. Remove the deflector from the flange only when replacement is necessary.



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48. Remove the breather pipe.



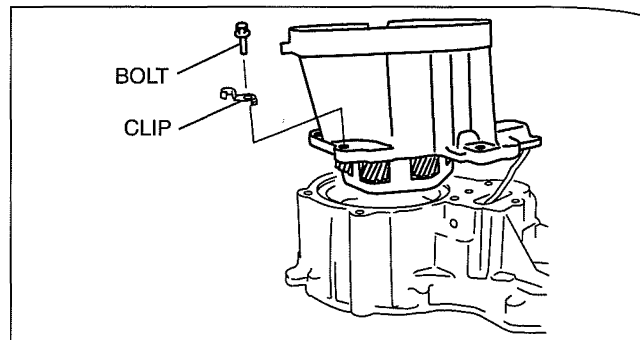
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TRANSFER [5R55S]

49. Remove the front transfer case six bolts.
50. Remove the front transfer case by separating the adapter sealer bond using a flathead screwdriver.

Caution

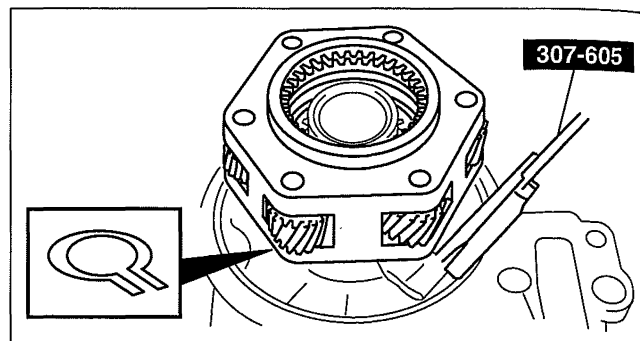
- When prying at the transfer case, be careful not to damage the case.



51. Remove the input shaft component and planetary gear component from the front transfer case.

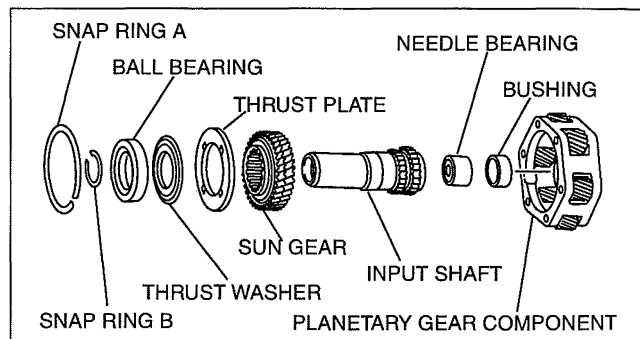
Note

- Expand the snap ring using the SST, and separate the planetary gear component and the input shaft component from the front transfer case.

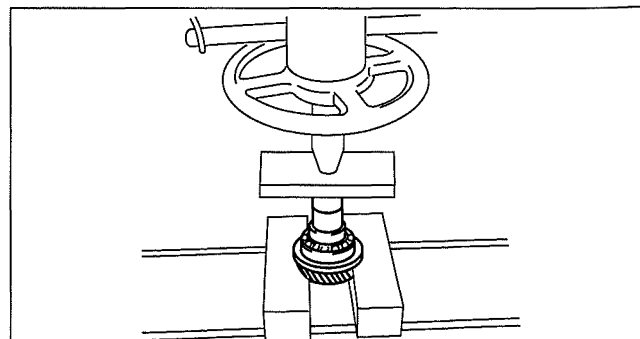


52. Remove the snap ring and oil seal from the front transfer case.

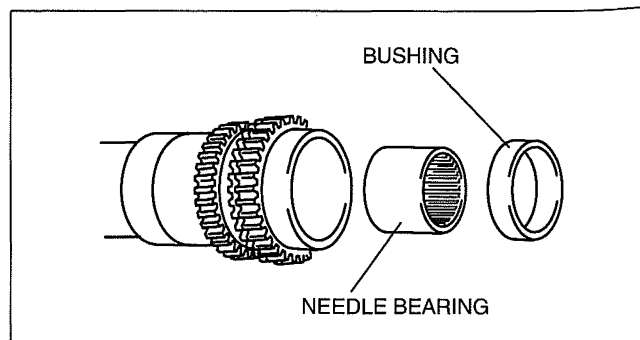
53. Remove the snap ring A.
54. Pull out the input shaft component and sun gear from the planetary gear component.
55. Remove the snap ring B.



56. Remove the ball bearing, thrust washer, thrust plate and from the input shaft using the press.

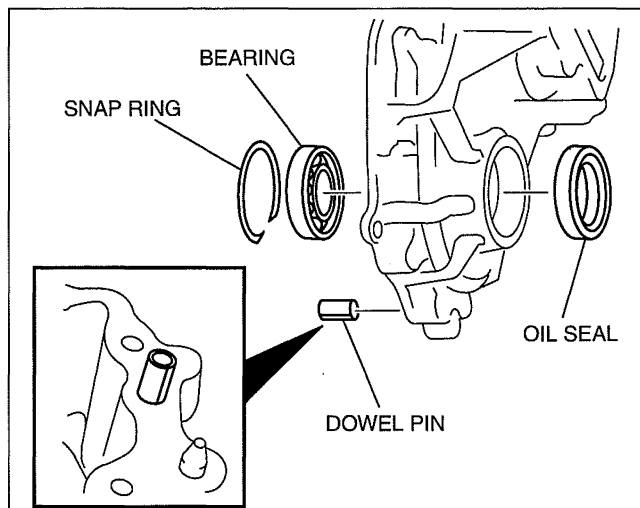


57. Remove the needle bearing and bushing from the input shaft component.



TRANSFER [5R55S]

58. Using a flathead screwdriver, remove the oil seal.
59. Remove the snap ring.
60. Remove the ball bearing.

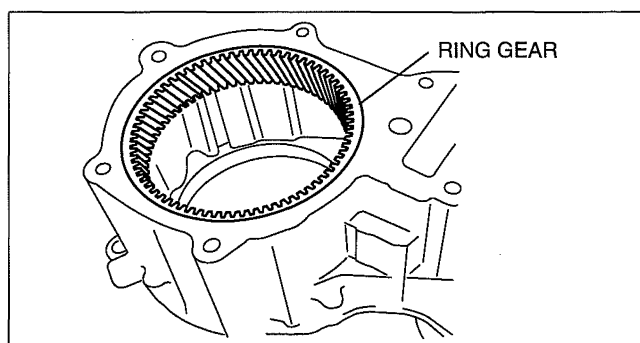


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61. Remove the dowel pins from the center transfer case.

Note

- Do not remove the ring gear from the center transfer case.



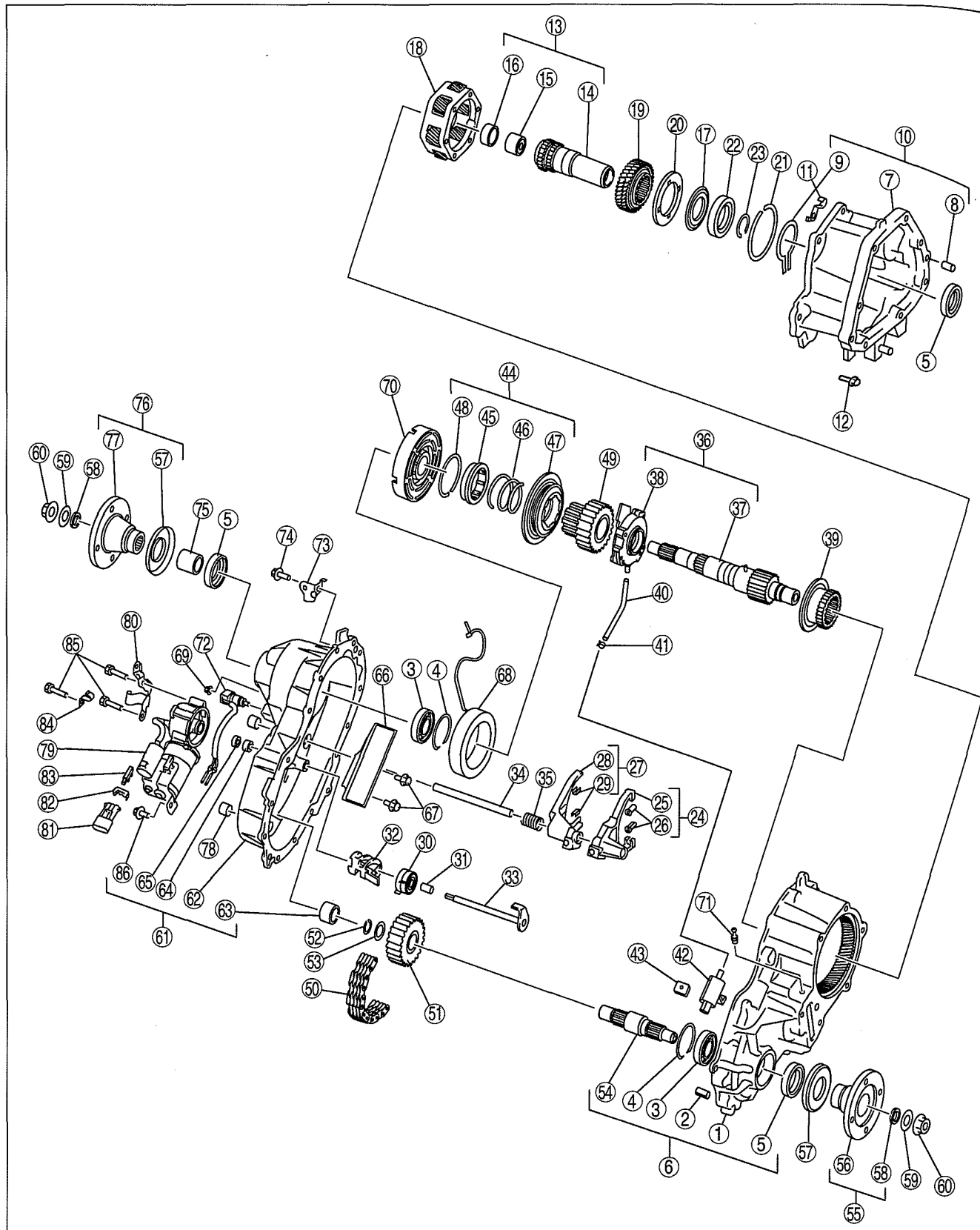
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TRANSFER [5R55S]

TRANSFER CASE ASSEMBLY[5R55S]

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Exploded View



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1	Center transfer case
2	Dowel pin
3	Bearing
4	Snap ring

5	Oil seal
6	Center transfer case component
7	Front transfer case
8	Dowel pin

TRANSFER [5R55S]

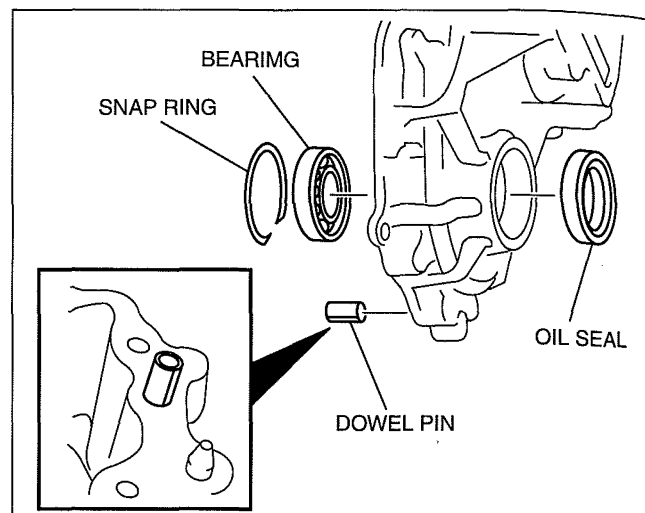
9	Snap ring
10	Front transfer case component
11	Clip
12	Bolt
13	Input shaft component
14	Input shaft
15	Needle bearing
16	Bushing
17	Thrust washer
18	Planetary gear component
19	Sun gear
20	Thrust plate
21	Snap ring A
22	Ball bearing
23	Snap ring B
24	Reduction shift fork component
25	Reduction shift fork
26	Reduction shift fork facing
27	Lockup shift fork component
28	Lockup shift fork
29	Lockup shift fork facing
30	Spring
31	Spacer
32	Cam
33	Shift shaft
34	Rail shift
35	Return spring
36	Output shaft and gerotor pump component
37	Output shaft component
38	Gerotor pump component
39	Reduction hub
40	Pump hose
41	Hose clamp
42	Oil strainer
43	Magnet
44	Lockup component
45	Lockup hub
46	Spring
47	Lockup collar

48	Snap ring
49	Drive sprocket
50	Drive chain
51	Driven sprocket
52	Snap ring
53	Spacer
54	Lower output shaft
55	Front output flange component
56	Front output flange
57	Deflector
58	Oil seal
59	Washer
60	Locknut
61	Rear transfer case component
62	Rear transfer case
63	Needle bearing
64	Sleeve
65	Seal
66	Snubber
67	Bolt
68	Clutch coil
69	Nut
70	Clutch housing
71	Breather pipe
72	Speed sensor component
73	Bracket
74	Bolt
75	Spacer
76	Rear companion flange component
77	Rear companion flange
78	Oil plug
79	Motor component
80	Bracket
81	Terminal connector
82	Clip
83	Bracket
84	Clip
85	Bolt
86	Bolt

TRANSFER [5R55S]

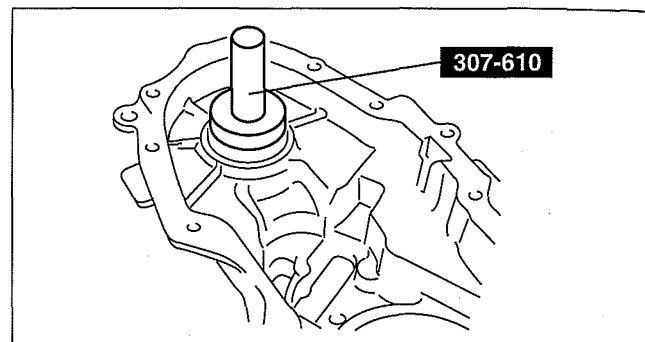
Assembly Procedure

1. Insert two new dowel pins.



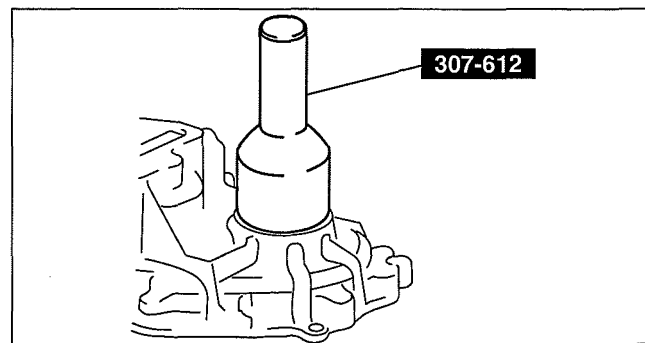
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2. Press the ball bearing into the center transfer case using the **SST**.
3. Install the snap ring.



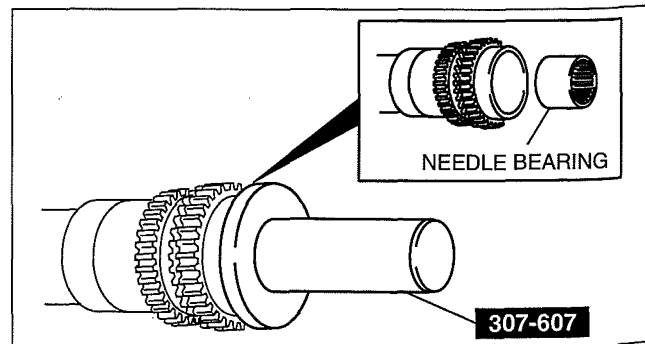
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4. Install the new oil seal, by pressing it into the center transfer case using the **SST**.



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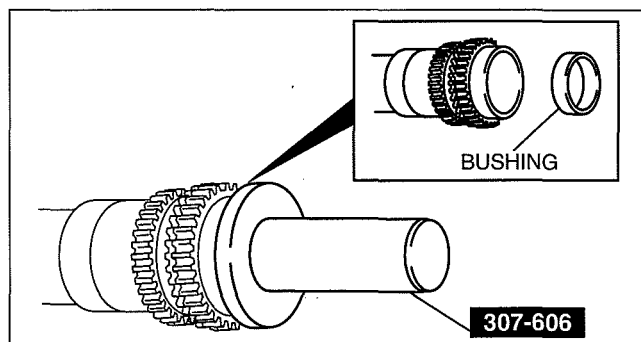
5. Press the new needle bearing into the input shaft using the **SST**.



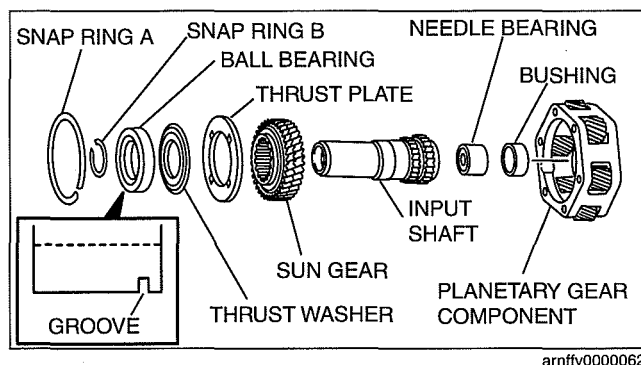
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TRANSFER [5R55S]

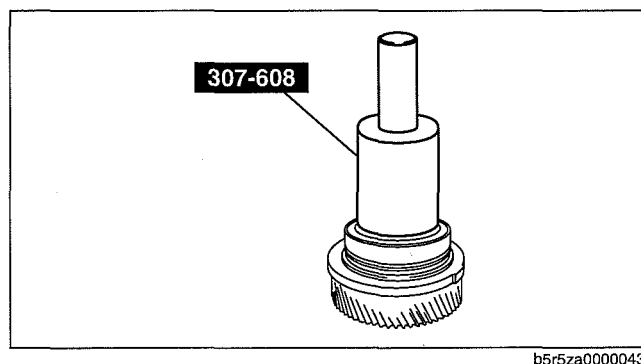
6. Press the new bushing into the input shaft using the **SST**.



7. Assemble the sun gear and thrust plate to the input shaft.



8. Assemble the thrust washer and ball bearing to the input shaft using the **SST**.
9. Install the snap ring B.

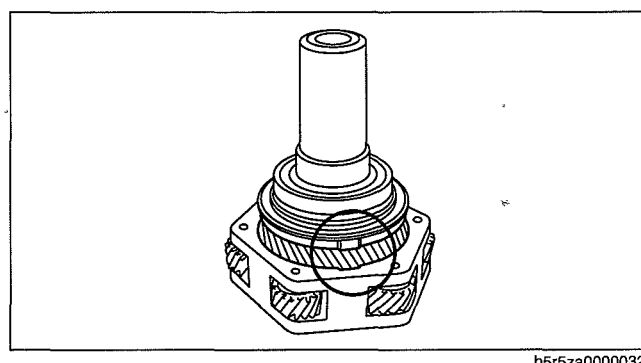


10. Assemble the planetary gear component to the planet carrier as shown in the figure.

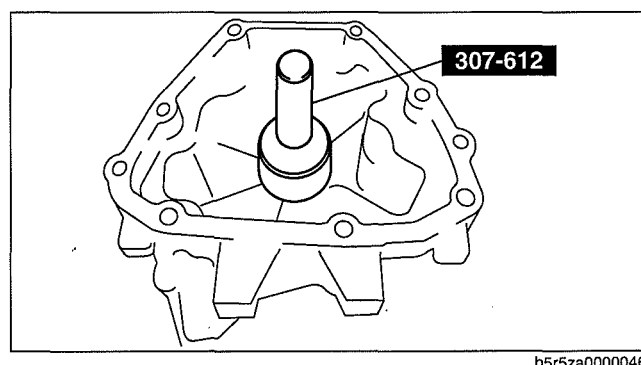
Note

- Verify that the retaining ring is assembled to the upper groove of the planetary gear component on the workbench.

11. Install the snap ring A to the planetary gear component.

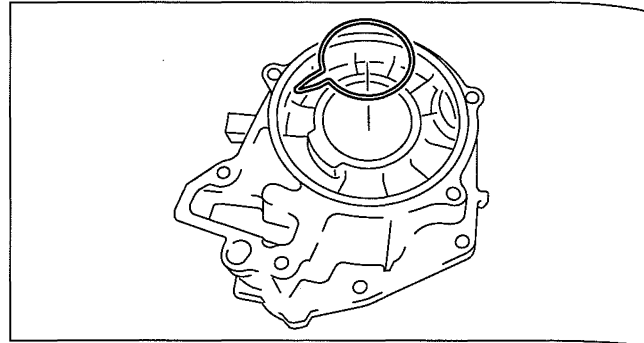


12. Press the new oil seal into the front transfer case using the **SST**.



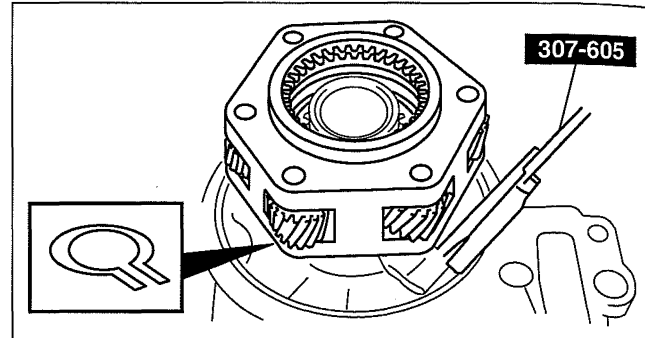
TRANSFER [5R55S]

13. Install the snap ring by making sure that the snap ring is correctly installed into the groove.



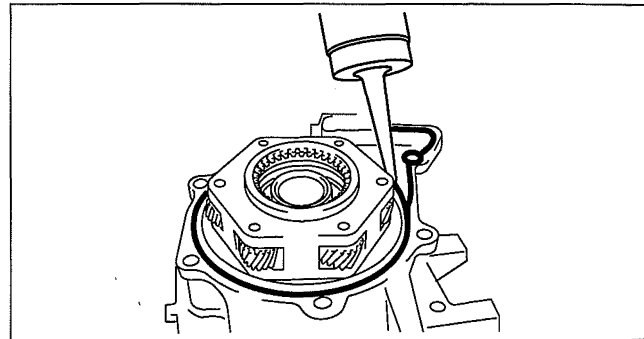
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14. Place the input shaft component onto the front transfer case, and assemble the planetary gear component to the groove while expanding the ends of the snap ring using the SST.



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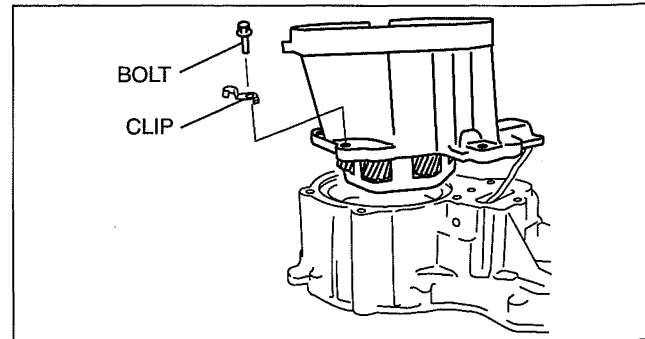
15. Apply sealant to the front transfer case as shown in the figure.



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16. Install the front transfer case on the center transfer case.
17. Install the six bolts and clip.

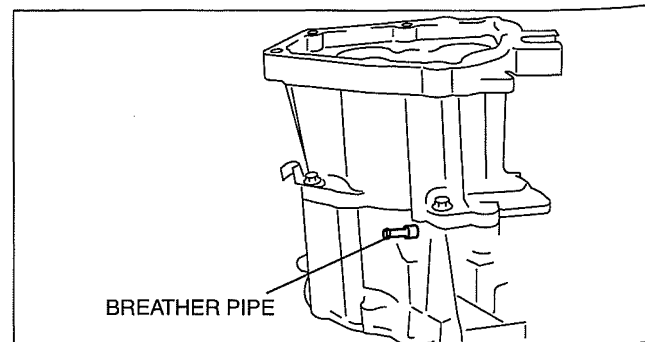
Tightening torque
34—41 N·m {3.5—4.1 kgf·m, 26—30 ft·lbf}



arnffv00000620

18. Install the breather pipe on the center transfer case.

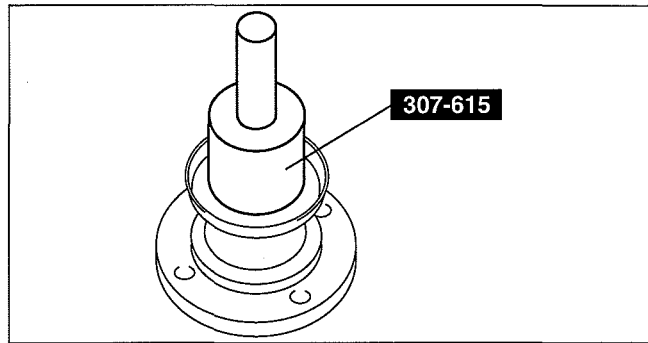
Tightening torque
4—7 N·m {41—71 kgf·cm, 36—61 in·lbf}



b5r5za00000422

TRANSFER [5R55S]

19. Press in the new deflector to the flange using the SST.

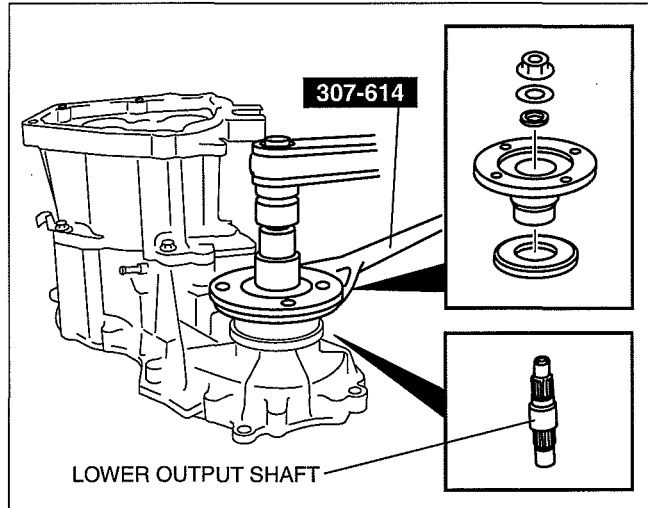


arnffv00000597

20. Assemble the lower output shaft to the center transfer case.
 21. Assemble the front output flange component, oil seal, washer, and nut.
 22. Secure the flange using the SST.
 23. Tighten the nut.

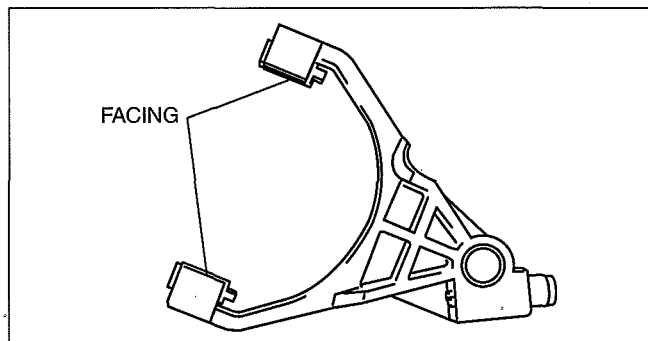
Tightening torque

305—332 N·m {31.2—33.8 kgf·m, 225—244 ft·lbf}



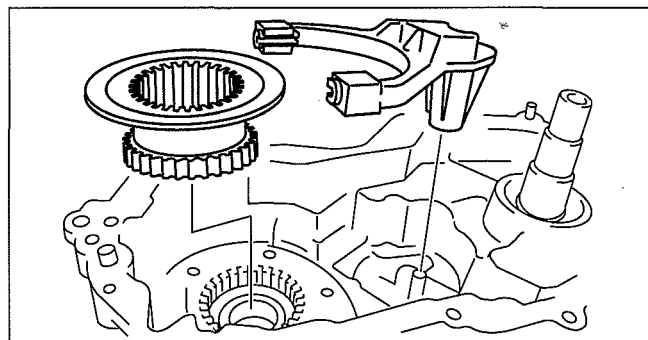
b5r5za00000466

24. Install the two-fork facing on the reduction shift fork component.



b5r5za00000438

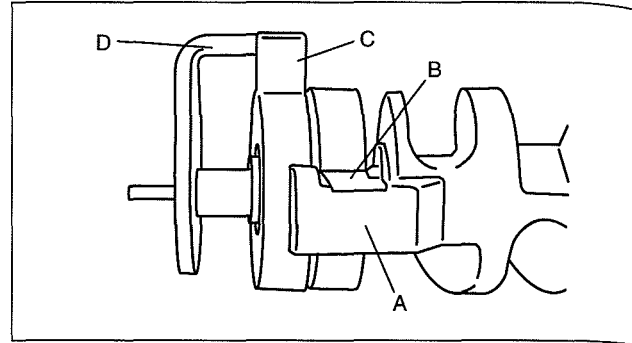
25. Install the reduction hub to the shift fork.
 26. Install the reduction hub and shift fork component to the planetary gear component.



b5r5za00000336

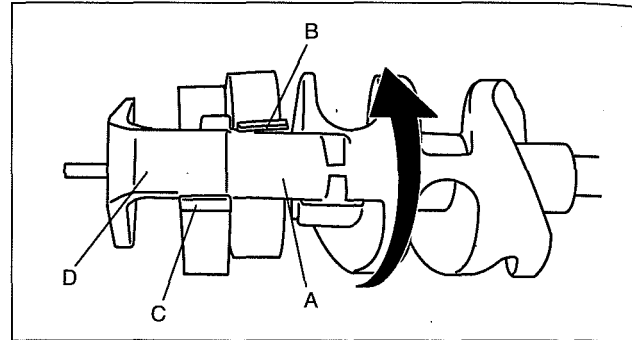
TRANSFER [5R55S]

27. Insert the spacer torsion spring and cam into the shift shaft.



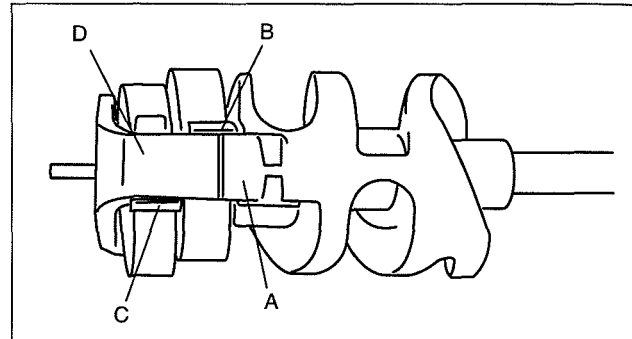
b5r5za000004

28. Press them in until area A of the cam contacts area B of the torsion spring, and area C of the torsion spring contacts area D of the shift shaft.



b5r5za000004

29. Rotate the cam to the position indicated in the figure, and press it to the shift shaft side until it contacts the torsion spring.

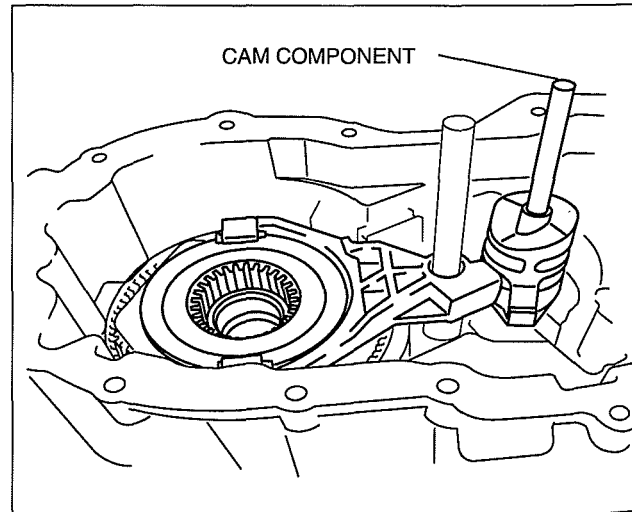


b5r5za000004

30. Install the cam component in the case as shown in the picture.

31. Verify that the reduction fork roller is seated in the cam groove as shown in the figure.

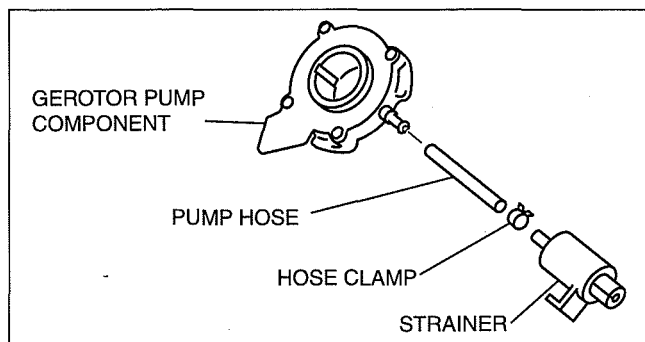
32. Assemble the rail shift to the reduction fork bore.



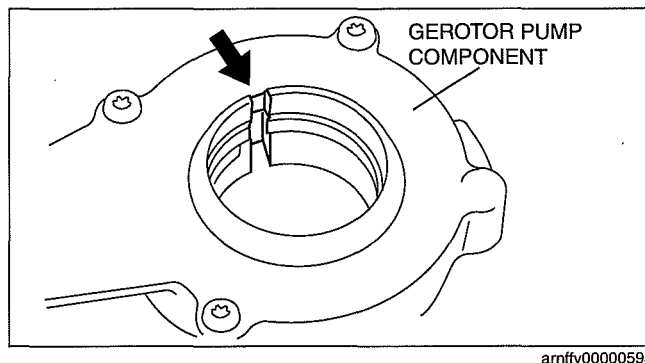
b5r5za000004

TRANSFER [5R55S]

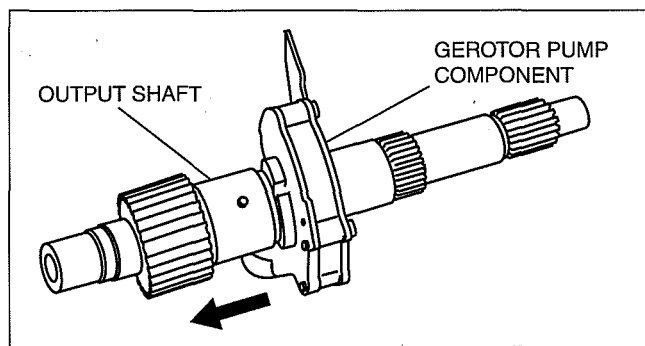
33. Using a hose clamp, tighten the pump hose at the end where it is coupled with the strainer.



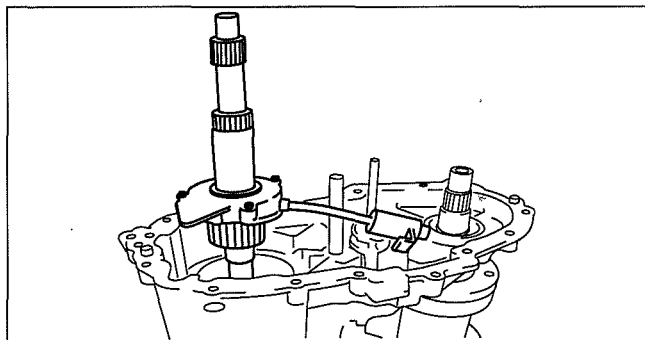
34. Align the gerotor pump rotor slot and the pump body slot so that they are in a single line.



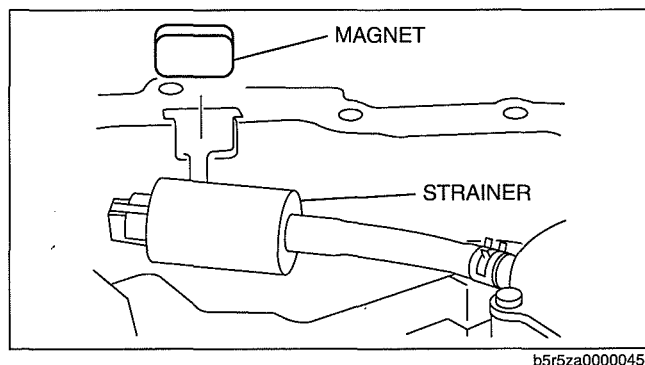
35. Slide the gerotor pump component on the output shaft over pump pin.



36. Assemble the output shaft from the center transfer case.
37. Assemble the output shaft spline to the reduction hub.
38. Engage the output shaft end with the input shaft bearing.

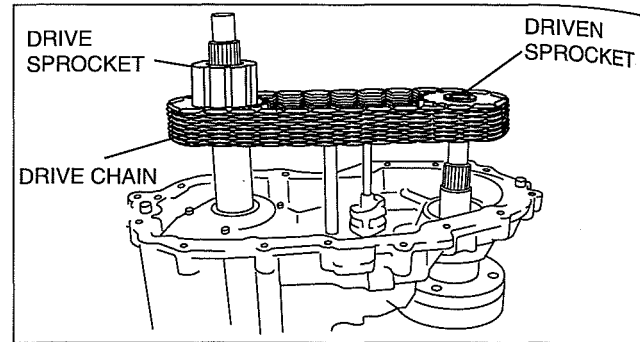


39. Couple the strainer with the case and insert the magnet into the transfer case slot.
40. Assemble the drive sprocket to the output shaft.
41. Assemble the lower output shaft to the driven sprocket.

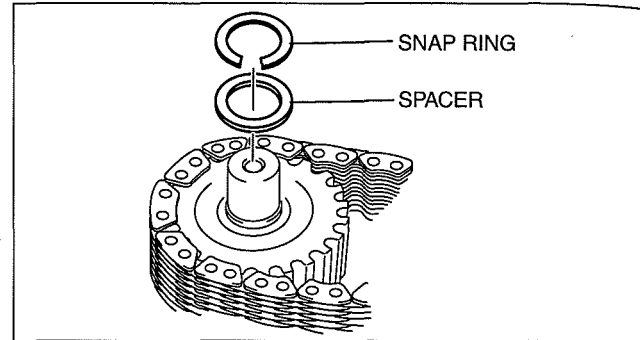


TRANSFER [5R55S]

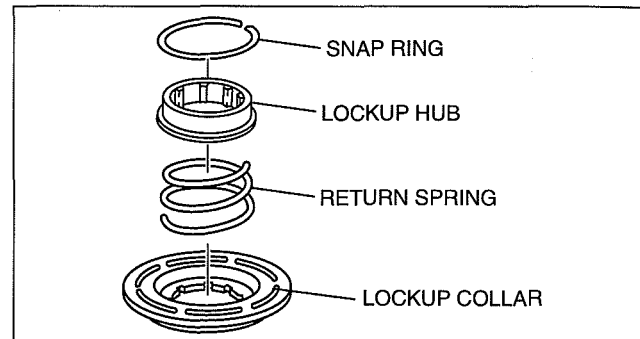
- 42. Install the drive chain onto the sprockets.
- 43. While keeping the tension of the drive chain, assemble the transfer case and drive chain component in parallel.
- 44. While rotating the driven sprocket, engage it with the front output shaft spline.



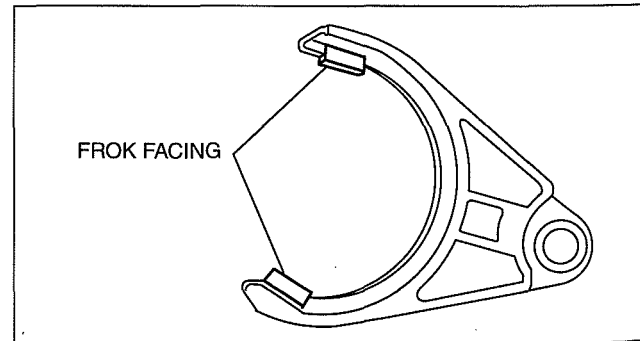
- 45. Assemble the spacer to the front output shaft.
- 46. Assemble the snap ring into the groove.



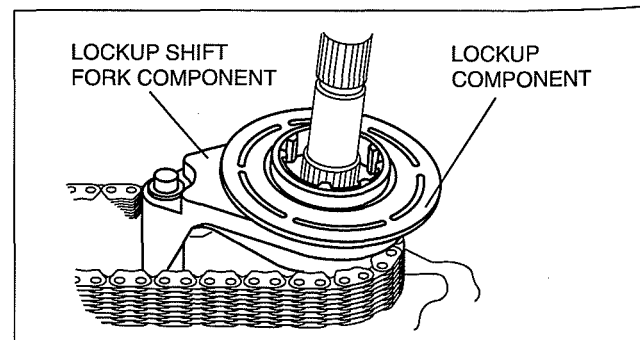
- 47. Install the lockup hub and return spring to the lock up collar and insert the snap ring.



- 48. Install the two new facings to the fork.

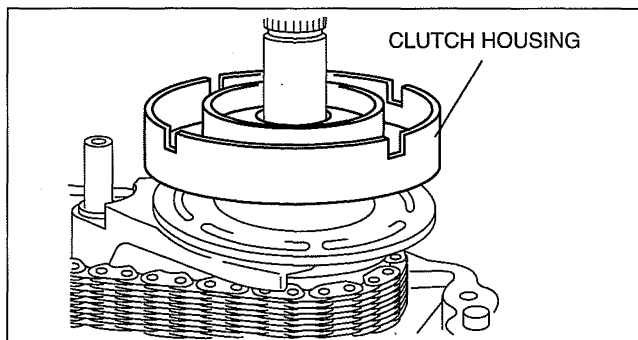


- 49. Assemble the lockup component to the lockup fork, and assemble them to the drive shaft and the rail shaft.



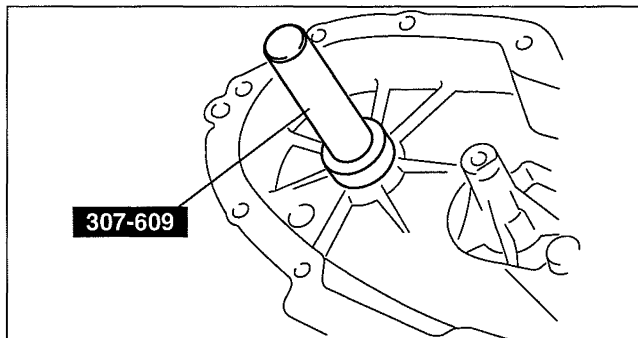
TRANSFER [5R55S]

50. Install the clutch housing on the output shaft.



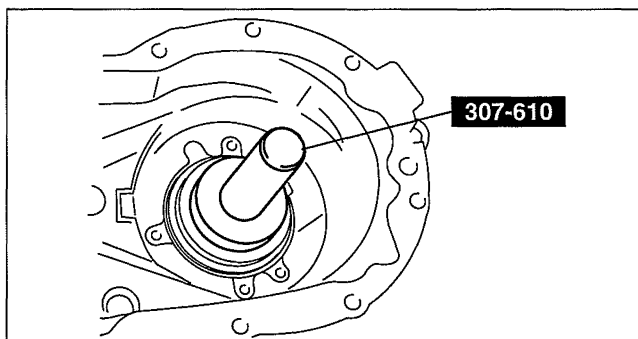
b5r5za00000312

51. Position the end of the new needle bearing with the identification mark facing upwards and press it into the cover using the **SST**.



b5r5za00000442

52. Press the ball bearing into the rear transfer case using the **SST** and install the snap ring.



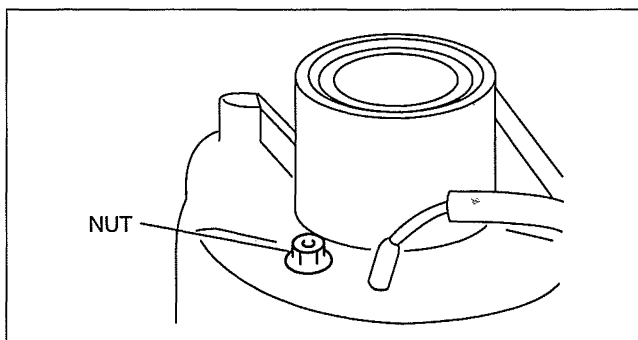
b5r5za00000443

53. Install the coil inside the rear transfer case.

54. Install the three nuts.

Tightening torque

8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



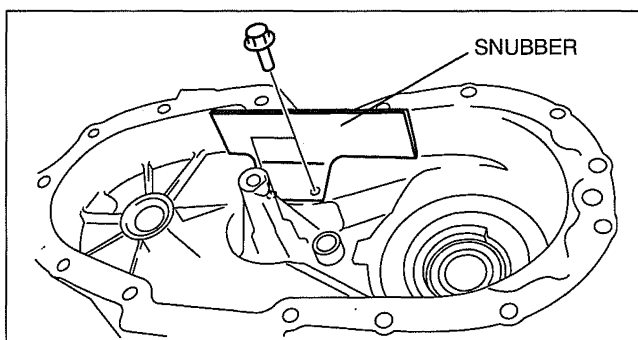
b5r5za00000310

55. Install the snubber.

Tightening torque

8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

56. Install the return spring over rail shaft in the rear transfer case.



b5r5za00000412

TRANSFER [5R55S]

57. Apply a 1.6 mm {0.063 in} bead of sealant to the transfer case mounting surface, and assemble the cover and the transfer case.

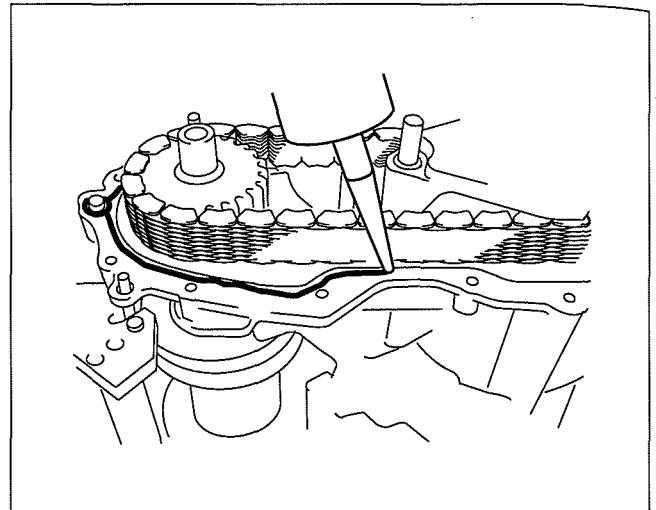
Caution

- Do not use excessive force.

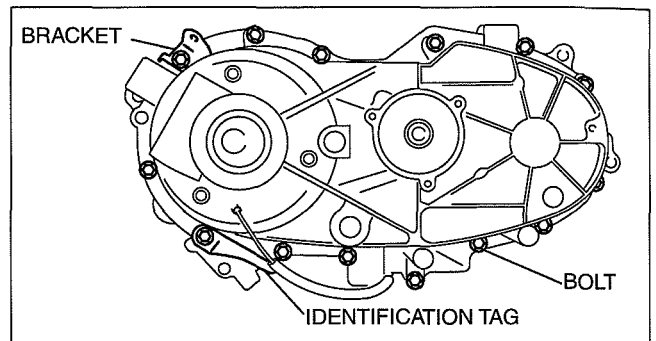
Note

- For installation of cover, align the cover with transfer case.

58. Install the rear transfer case onto the center transfer case as follows.
- (1) Align the cover holes with the transfer case dowel pins.
 - (2) Align the cover bearings with output shafts.
 - (3) Align the shift shaft with cover boss.
 - (4) Align the cover blind hole with rail shaft and make sure that return spring is not cocked.
59. Install the 14 bolts identification tag and bracket.

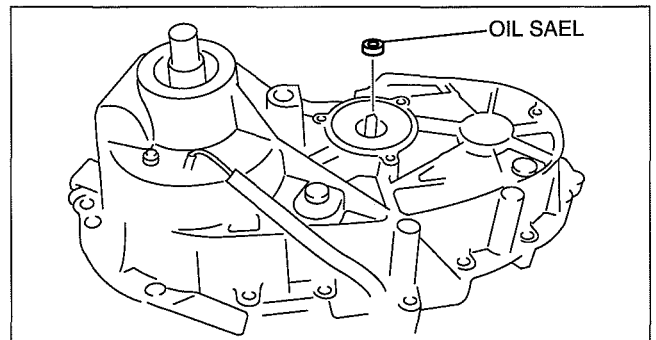


b5r5za00000350



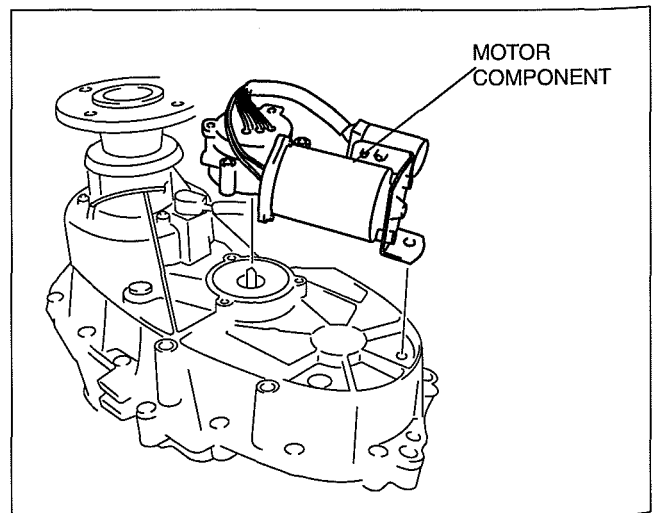
b5r5za00000429

60. Install the new oil seal.



arnfv00000558

61. Align the motor component with the shift shaft and assemble it to the rear transfer case.
62. Rotate the motor in clockwise direction to check correct engagement.



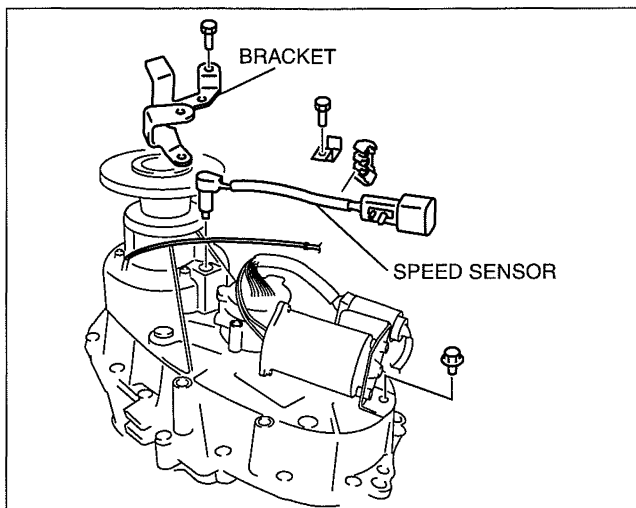
arnfv00000615

TRANSFER [5R55S]

63. Install the speed sensor in the rear transfer case.
64. Install the bracket and bolts to the motor component.

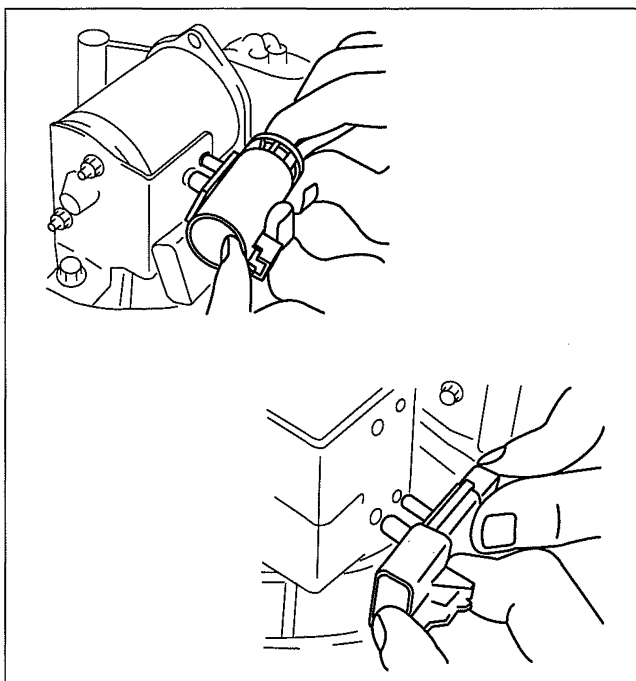
Tightening torque

8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



arnffv00000622

65. Pass the clutch coil wiring harness through the sensor harness sleeve, and connect the coil terminal to the connector.
66. Install the motor connector and sensor connector to the motor bracket.

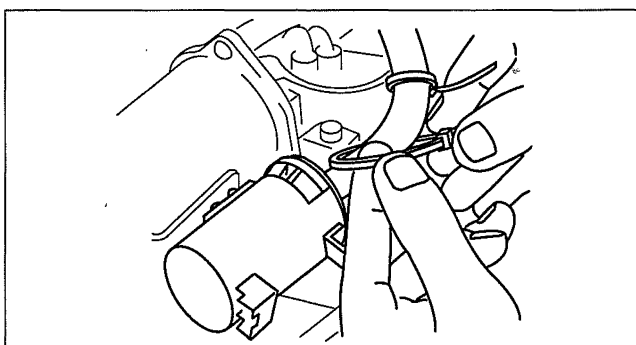


b5r5za00000352

67. Bundle the wiring harness between the speedometer component and terminal connector using a wiring harness protector and tie wraps.
68. Install the oil plug to the rear transfer case.

Tightening torque

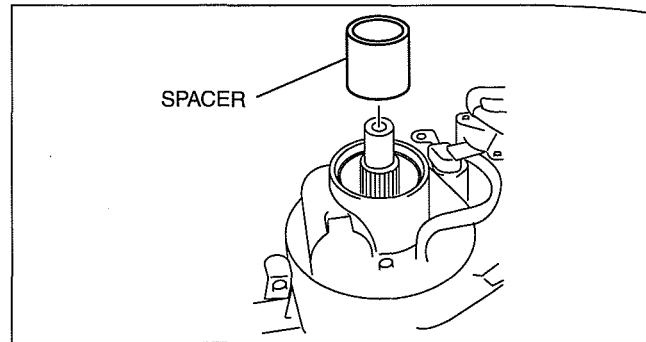
19—30 N·m {2.0—3.0 kgf·m, 15—22 ft·lbf}



b5r5za00000353

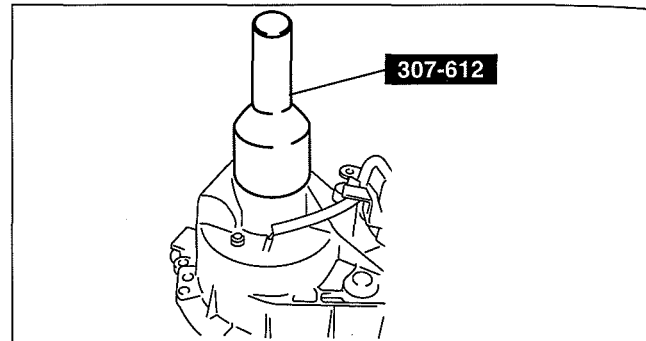
TRANSFER [5R55S]

69. Install the spacer over output shaft spline.



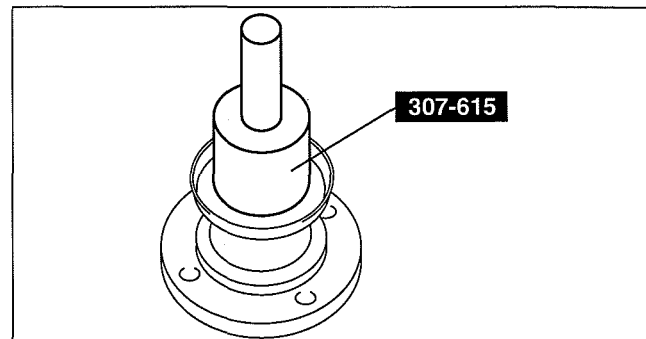
b5r5za00000454

70. Press the new oil seal into the rear transfer case component using the **SST**.



b5r5za00000445

71. Press in the new deflector to the flange using the **SST**.



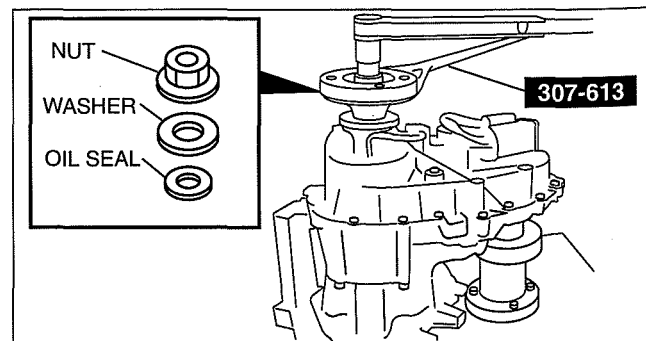
arnffv00000597

72. Install the rear companion flange component, oil seal, washer and nut.

73. Holding the rear companion flange using the **SST**, tighten the nut.

Tightening torque

305—332 N·m {31.2—33.8 kgf·cm, 225—244 in·lbf}



b5r5za00000355

TRANSFER CASE INSPECTION[5R55S]

id0316c1501300

Note

- Always replace the hose coupling, O-ring and oil seal with new parts.

1. Visually check all the parts for damage.
2. Referring to normal gear tooth face, specifically inspect the uneven wear and chips of gear tooth, replace if necessary.

TECHNICAL DATA

03-50 TECHNICAL DATA

TECHNICAL DATA 03-50-1

TECHNICAL DATA

DCF03500000W01

03

Item	Specification
Maindrive gear bearing end play	00-15 mm {0.00-0.059 in}
Mainshaft center bearing end play	00-15 mm {0.00-0.059 in}

Maindrive gear bearing retaining ring

Thickness (mm {in})
2.60 {0.102}
2.67 {0.105}
2.74 {0.108}

Countershaft adjustment shim selective chart

Dimension A (mm {in})	Shim thickness (mm {in})
25.025.1 {0.9850-988}	0.5 {0.020}
25.125.2 {0.9890-992}	0.4 {0.016}
25.225.3 {0.9930-996}	0.3 {0.012}
25.325.4 {0.9974-000}	0.2 {0.008}
25.425.5 {1.0004-003}	0.1 {0.004}
25.525.6 {1.0044-007}	-

Mainshaft center bearing retaining ring

Thickness (mm {in})
3.1 {0.122}
3.2 {0.126}
3.3 {0.130}



SERVICE TOOLS

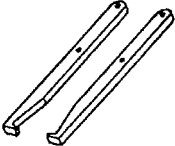
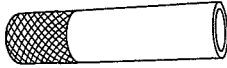
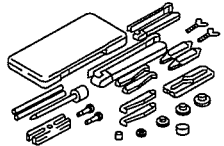
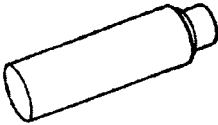
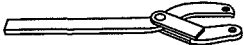
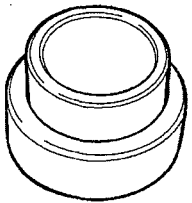
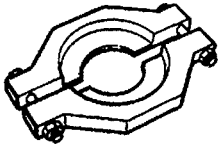
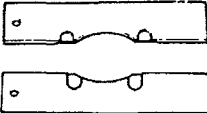
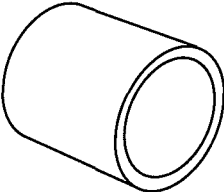
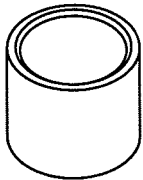
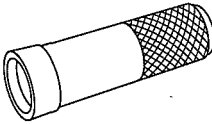
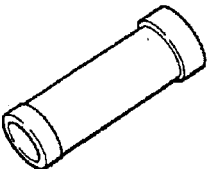
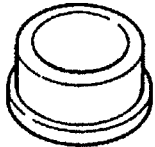
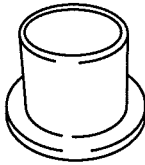
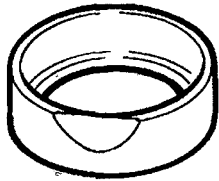
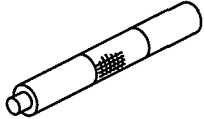
03-60 SERVICE TOOLS

SERVICE TOOLS 03-60-1

SERVICE TOOLS

DCF03600000W01

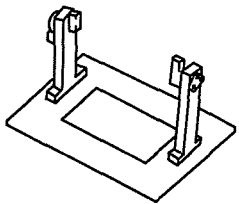
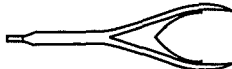
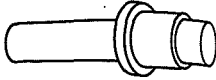
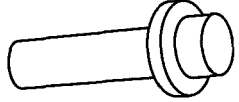
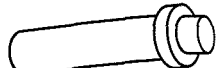
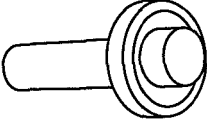
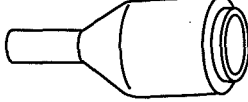
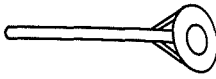
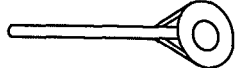
03

49 H017 101 Hook 	49 0727 415 Bearing Installer 	49 0839 425C Bearing puller set 
49 E032 303 Bearing Remover 	49 S120 710 Coupling flange holder 	49 F401 336B Attachment 
49 H027 002 Bearing remover 	49 F026 103 Wheel Hub Puller 	49 G028 203 Support 
49 H001 628 Guide Pin 	49 F401 331 Body 	49 U025 001 Installer Protector 
49 F027 009 Attachment 68 And 77 	49 T019 005 Support Block 	49 G033 107A Installer Dust Cover 
49 M005 797 Installer oil seal 		

SERVICE TOOLS [5R55S]

SERVICE TOOLS[5R55S]

id0360e1500100

307-611 Repair fixture 	307-605 Snap ring pliers-adapter 	307-607 Drift NRB input shaft 
307-606 Drift bush input shaft 	307-608 Drift ball bearing input shaft 	307-609 Drift NRB fitting cover 
307-610 Drift ball bearing 	307-612 Seal driver 	307-614 Flange holder front 
307-613 Flange holder rear 	307-615 Dust deflector press tool 